Dell PowerFlex 5.0.x and PowerFlex Manager 5.0.x

Install Guide

September 2026
Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Introduction

The *Dell PowerFlex 5.0.x and PowerFlex Manager 5.0.x Install Guide* describes how to deploy a new instance of PowerFlex 5.0.x.

Use this guide only for PowerFlex software installations. PowerFlex rack and PowerFlex appliance customers should not use this document for installations. For PowerFlex rack documentation, go to PowerFlex rack technical documentation. For PowerFlex appliance documentation, go to PowerFlex appliance documentation.

The target audience for this document includes Dell Technologies customers and services.

For additional PowerFlex software documentation, go to PowerFlex Software: Technical Documentation.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

Persistent memory

Persistent memory (SDPM/NVDIMM) is not mandatory for inline data compression, instead PMEM is used for write cache and metadata persistency in protection domain with erasure coding.

Persistent memory serves to extend the life time of SSD or NVMe disks, when write cache is used to provide faster acknowledge of IO, it also reduces the wear level on disks.

Persistent memory requires 64/128GiB when using SDPM or NVDIMM using for write cache.

Write cache (WrC)

Write cache (WrC) is a facility that provides high availability, low write latency, and aggregate writes to data stored in PMEM (such as SDPM or NVDIMM) or SSD. Write cache is a distributed component that provides high availability by writing to storage on multiple nodes. Updates are first written into the write cache.

We have two types of device groups:

- Storage devices
- Persistent memory (optional)

Read cache

PowerFlex PDS also implements Read Cache using available free memory to provide faster response time for reads.

This is sometimes referred to as MDCache

The PowerFlex MD Cache that are enabled in 5.0 retains frequently used metadata in the RAM of the storage node, improving read latency and IOPS by avoiding expensive read operations from the drives. The MD Cache also enhances the performance of small write operations (below 8 KB).

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Revision history

Table 2. Revision history

Date	Document revision	Description of changes
September 2026	1.1	Editorial updates
October 2025	1.0	Initial release

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Installation overview and prerequisites

Installation workflow

Installing PowerFlex on physical servers includes the following activities:

1. Complete the installation checklist to prepare your environment for PowerFlex Manager.

   For PowerFlex custom node, see the *Hardware Configuration and Operating System Installation Guide* available on the Dell Technologies support site. These documents provide step-by-step installation instructions on how to prepare the server and operating system before installing PowerFlex.

2. Install and configure PowerFlex Manager.
3. Install the PowerFlex components.
4. For a two-layer system configuration, install storage data clients (SDCs) and/or configure NVMe over TCP hosts on external servers.
5. Upload the PowerFlex license.
6. Perform any necessary post-installation tasks.
7. Move the PowerFlex system to production.

Supported configurations

The following PowerFlex Manager configurations are supported:

Table 3. PowerFlex Manager supported configurations

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (physical nodes) required	CPU resource requirement (per VM/ node)	Mempry resource requirement (per VM/node)	Hard disk requirement (per VM/node)
Pre-existing VMware ESXi management infrastructure	• PowerFlex appliance • PowerFlex custom node • PowerEdge servers • Non-Dell Technologies hardware	Customers may use their own VMware ESXi hypervisor to host the MVMs, provided that the underlying infrastructure meets the platform's minimum system requirements. PowerFlex management platform is hosted on the three MVMs. Use the installer to set up the PowerFlex management platform. The MVMs and installer VM are deployed using the MVM OVA and installer OVA resepctively. Download the OVA files from the Dell Technologies support site. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during	One installer VM or physical node for installer	8 vCPUs/ 8 CPUs	8 GB	100GB
			Three MVMs for installing	14 vCPUs	32GB	600GB

Table 3. PowerFlex Manager supported configurations (continued)

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (physical nodes) required	CPU resource requirement (per VM/ node)	Mempry resource requirement (per VM/node)	Hard disk requirement (per VM/node)
		the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.				
Integrated management (non-KVM) on PowerFlex	• PowerFlex custom node • PowerEdge servers • Non-Dell Technologies hardware	This is essentially a management controller that integrates with the PowerFlex software components on the same physical servers, instead of requiring a dedicated set of management nodes. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed. The bare-metal node is repurposed for the MDM role.	Three physical nodes for installing and configuring PowerFlex management platform	14 vCPUs	32GB	600GB
			One physical node or VM for the installer	8 vCPUs / 8 CPUs	8 GB	100GB
Pre-existing Linux bare metal / Linux VM management infrastructure	• PowerFlex custom node • PowerEdge servers • Non-Dell Technologies hardware	Customers may provide three bare-metal Linux servers to host the PowerFlex management platform. It is required to install the operating system and configure the network settings on each server. All hardware must comply with the minimum system requirements specified for the platform. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform. Once the	Three physical nodes for installing and configuring PowerFlex management platform	14 CPUs	32 GB	600GB
			One VM or physical node for installer	8 vCPUs/8 CPUs	8 GB	100GB

Table 3. PowerFlex Manager supported configurations (continued)

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (physical nodes) required	CPU resource requirement (per VM/ node)	Mempry resource requirement (per VM/node)	Hard disk requirement (per VM/node)
		deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.				
Pre-existing Linux KVM management infrastructure	• PowerFlex custom node • PowerFlex appliance • PowerEdge servers • Non-Dell technologies hardware	Customers may use their own Linux KVM hypervisor to host the management virtual machines, provided that the underlying infrastructure meets the platform's minimum system requirements. Download the QCOW files from the Dell Technologies support site. **NOTE:** The installer can be deployed either on a virtual machine or a Linux bare metal server. It is required only during the initial setup of the PowerFlex management platform. Once the deployment is complete, the installer VM or the Linux bare metal server configuration can be safely removed.	Three MVMs for installing and configuring PowerFlex management platform	14 vCPUs	32 GB	600GB
			One VM or physical node for installer	8 vCPUs/8 CPUs	8GB	100 GB

Installation checklist

This section contains a checklist of items required for installing PowerFlex. Ensure that you have done everything on this list before proceeding then perform the remaining preparation tasks covered in this guide.

Table 4. PowerFlex installation checklist

Item	Checked
General	
All servers intended for hosting PowerFlex components and PowerFlex Manager must meet system requirements. **NOTE:** The Dell-supplied hardware configurations must satisfy all the hardware requirements.	
Secure boot on servers is not supported. Ensure that secure boot is disabled in the BIOS of each server.	
All nodes must have unique hostnames.	
To use secure authentication mode, ensure that the latest OpenSSL 64-bit 1.0.2 or above is installed on all servers in the system.	
Ensure that you know the following: • Primary and secondary MDM IP addresses	

Table 4. PowerFlex installation checklist (continued)

Item	Checked
• Primary MDM admin username and password • Primary MDM operating system hostname and password • LIA password	
VMware ESXi	
Ensure that the required network interfaces and IP addresses are configured.	
Networking	
For more information on networking requirements, see the *Dell PowerFlex Technical Overview*.	
Powerflex Manager and its installer node	
The nodes that host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. All management nodes should be on the same subnet and SSH should be enabled.	
Prepare the following nodes for installing PowerFlex Manager • PowerFlex Manager installer node: ○ If the PowerFlex Manager installer is on a bare-metal node, install Docker version 20.10.0 or higher (Podman version 4.9.4 is also supported). ○ If you are using PowerFlex Manager installer on an existing VMware ESXi management infrastructure, deploy the provided installer OVA (SUSE Linux-based). • If you are using the PowerFlex Manager installer on existing KVM management infrastructure, deploy the provided installer QCOW2 (SUSE Linux-based). • Three management nodes: ○ These nodes must have IP addresses and hostnames that are unique and are on the same subnet. ○ Hostname labels may contain only the ASCII letters 'a' through 'z' (in a case-insensitive manner), the digits '0' through '9', and the hyphen '-'. ○ The PowerFlex management platform can be deployed in the following modes: ▪ One dedicated management node. ▪ Three dedicated management nodes with VMware ESXi or bring your own Linux dedicated management nodes. ▪ Integrated PowerFlex management (non-KVM) with storage nodes.	

Deploying PowerFlex Manager

Deploy PowerFlex Manager in accordance with your environment.

PowerFlex Manager can be installed in three ways:

- VMware ESXi environment: Three VMs are running a management virtual machine cluster. PowerFlex Manager is deployed using a fourth VM acting as an installer VM.
- Integrated PowerFlex management (non-KVM): PowerFlex Manager is installed on three bare metal nodes, and then PowerFlex block storage is deployed on the same nodes.
- Linux management environment: The PowerFlex management platform is installed on the Linux operating system directly.
- PowerFlex management cluster (KVM): Enables the setup of the PowerFlex management platform system for management purposes in a Kubernetes VM environment.

Software packages

For more information about software packages, see the *Dell PowerFlex 5.0.x Release Notes* available on the Dell Technologies support site.

Download PowerFlex installation packages

Use the following procedure to download and extract the installation bundle if you are deploying a PowerFlex cluster outside of PowerFlex Manager.

About this task

Based on the deployment scenario, download either of the following PowerFlex installation packages:

- Use the following packages for vSphere, integrated PowerFlex management (non-KVM), and Linux deployments:
  - `Software_Only_Complete_5.0.x.x.zip`— Contains core PowerFlex software and the compliance matrix (`cm-xxxxxxxx.gpg`) file
- Use the following package for VMware vSphere or virtualized deployments:
  - `PowerFlex_Manager_5.0.x.x.zip`— Contains the OVA templates for the Installer node and for the management node
- Use the following package for deploying with a bare metal Linux installation:
  - `PFMP2-5.0.x.x-xxx.tgz`— Contains the files and scripts for PowerFlex Manager deployment (bare-metal)

Steps

1. On the Dell Technologies support site, download the installation bundle for the version of PowerFlex.
2. Copy the installation bundle to your personal computer or desktop.
3. Extract the installation bundle.

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Deploying the PowerFlex management platform

To deploy the PowerFlex management platform, use one of the following installation options:

- Deploying the PowerFlex management platforminto a VMware vSphere environment
- Deploying the PowerFlex management platforminto a Integrated PowerFlex management (non-KVM) environment
- Deploying the PowerFlex management platforminto a Linux management environment
- Deploying the PowerFlex management platforminto a PowerFlex management cluster (KVM) environment

Deploying the PowerFlex management platform in a VMware vSphere environment

Use this procedure to deploy the PowerFlex management platform in a VMware vSphere environment.

PowerFlex management platform virtual machine requirements

Before you begin, ensure that the VMs you are installing the PowerFlex management platform meet the following requirements:

The PowerFlex management platform virtual machine requirements for storage, CPU, and RAM are the following:

- PowerFlex version: 5.0.x

Table 5. VM requirements

System scale	CPU cores	RAM (GB)	Storage capacity (GB)
Low Scale	10	32	300
Medium Scale	10	32	350
High Scale	14	32	450

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This section includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces that must be configured:

NOTE: For PowerFlex, including the `oob` VLAN is optional.

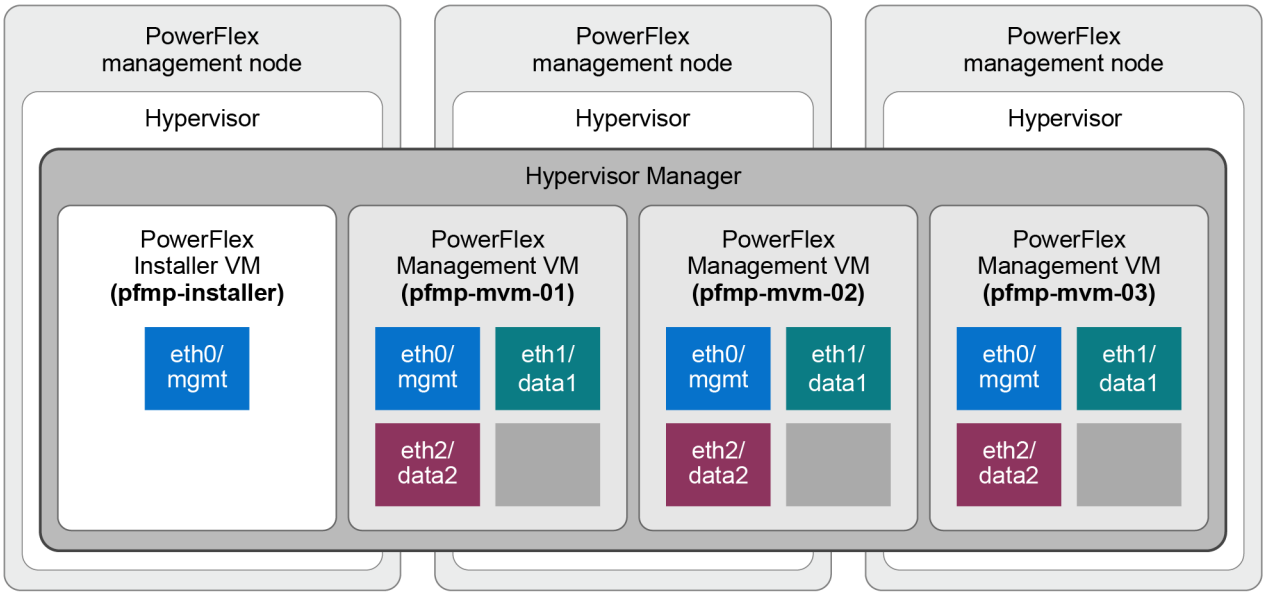


Figure 1. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploying and configuring the PowerFlex management platform installer virtual machine

Add Entrust root certificate to the VMware vCenter certificate store

If you want to use third-party certificates in your environment, you must add a third-party trusted root certificate to the VMware vCenter certificate store.

Steps

1. Manually download the OVCS2 certificate (OVCS2-CSBR1-crosscert) from the Entrust website: Entrust Certificate Services Subordinate CAs.

   Search for the keyword **OVCS2**. Click **CRT** to download the certificate.
2. Upload the Entrust root certificate to the VMware vCenter certificate store:
   a. Log in to the vCenter Server.
   b. Click **Administration** > **Certificates** > **Certificate Management**.
   c. Next to **Trusted Root Store**, click **Add**.
   d. Browse and select the downloaded Entrust root certificate. The certificate is added in a panel under **Trusted Root Certificates**.
3. Once the certificates are installed, OVA can be verified and deployed.

Deploy the PowerFlex management platform installer VM

Use this procedure to deploy and configure the PowerFlex management platform installer VM in a VMware environment.

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA Template**
5. Click **Open** > **Next**.
6. Enter **pfmp-installer** for the VM name.

7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Select the **PFMP datastore**.
11. Select **Virtual disk format** > **Thin provision** > **Next**.
12. Select **flex-node-mgmt-<vlanid>** > **OK** > **Next**.
13. Click **Finish**.
14. Right-click the VM and select **Power** > **Power On**.

Configure the PowerFlex management platform installer networking interface

Use this procedure to configure the networking for the PowerFlex management platform installer VM.

About this task

The following network adapter needs to be configured on the Powerflex management platform installer:

VLAN 105: flex-node-mgmt (eth0)

Steps

1. Launch the web console from vCSA and log in as **delladmin**.
2. To configure the <flex-node-mgmt> interface, type **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

Example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

a. To configure the default route, type:
sudo vi /etc/sysconfig/network/routes
default <gateway ip> - <interface>
Example: **default 10.10.105.1 - eth0**

b. To configure DNS search and DNS servers, type **sudo vi /etc/sysconfig/network/config** and modify the following: **NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>" NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"**
Example: **NETCONFIG_DNS_STATIC_SEARCHLIST="example.com" NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"**

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs.

Steps

1. To configure the NTP client, enter the following:
server <ntp ip address> iburst
For example: **server 10.10.105.240 iburst**
2. To start the chronyd service, enter:

sudo systemctl start chronyd

3. To enable chronyd, enter:
   sudo systemctl enable chronyd
4. To restart chronyd, enter:
   sudo systemctl restart chronyd
5. To verify that the network time protocol service is running, enter:
   sudo systemctl status chronyd.service
6. To check that the server is synced to NTP, enter:
   chronyc tracking

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status

   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

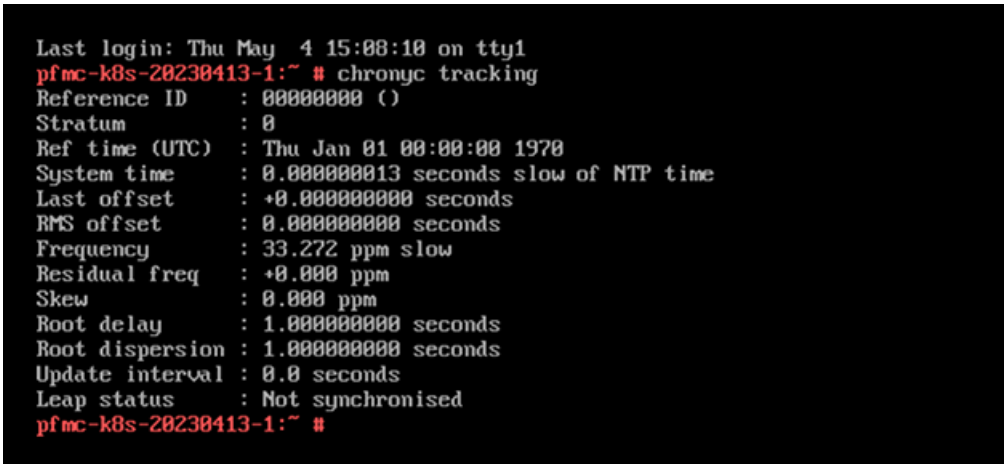
```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
System time : 0.000531569 seconds slow of NTP time
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

   Unsuccessful chronyc tracking status

   - `Stratum` is 0 (not synced).
   - Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

   The following image describes an unsuccessful chronyc tracking status.



7. Change the hostname, type the following: **`sudo hostnamectl set-hostname <mvm hostname>`**.

Deploying and configuring the PowerFlex management platform VMs

Use the following procedure to deploy the PowerFlex management platform cluster VMs.

Prerequisites

The PowerFlex management platform cluster requires three VMs to be deployed.

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA**.
5. Click **Open** > **Next**.
6. Enter **`pfmp-mvm-<number>`** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Review details and click **Next**.
11. Select the datastore.
12. Select **Virtual disk format** > **Thin provision** > **Next**.
13. Select **Desired Destination Network** > **OK** > **Next**.
14. Click **Finish**.
15. Repeat all the steps for the three management VMs.

Configure the PowerFlex management platform networking interfaces

Use the following procedure to configure the networking files for the PowerFlex management platform cluster VMs.

Prerequisites

NOTE: Based on your requirement, configure only the required interfaces on the management VMs. If you have unused interfaces on the deployed management VMs, then ensure you disconnect those interfaces within the vCenter, for example, an out-of-band interface. If the unused interfaces are connected and unconfigured, then PowerFlex Manager may display unexpected errors.

Steps

1. Launch the web console from the Virtual Machine Manager, log in as **`delladmin`**.

2. Configure the flex-node-mgmt-<vlanid> eth0 interface:
   a. Enter: **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.30
NETMASK=255.255.255.0
```

   b. Configure the default route.

   Enter **sudo vi /etc/sysconfig/network/routes**

   default <gateway ip> - <interface>

   For example:

   default 10.10.105.1 - eth0

   c. Configure DNS search and DNS servers.

   Enter **sudo vi /etc/sysconfig/network/config** and modify the following:

   NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

   NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

   For example:

   NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

   NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"

   d. To save and close, press **ESC** + **:wq**.
3. Configure the flex-data1-<vlanid> eth1 interface:
   a. Enter **sudo vi /etc/sysconfig/network/ifcfg-eth1**.

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.151.30
NETMASK=255.255.255.0
```

4. Configure the flex-data1-<vlanid> eth2 interface:
   a. Enter: **sudo vi /etc/sysconfig/network/ifcfg-eth2**.

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.152.30
NETMASK=255.255.255.0
```

5. Run the following command to update the network configuration:
   `sudo systemctl restart network`
6. Run the following command to change the delladmin account password:
   `passwd delladmin`
7. Change the root password with the following command:
   `sudo passwd root`
8. Repeat all steps for all PowerFlex management platform VMs.

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs.

Steps

1. To configure the NTP client, enter the following:
   `server <ntp ip address> iburst`
   For example: **`server 10.10.105.240 iburst`**
2. To start the chronyd service, enter:
   `sudo systemctl start chronyd`
3. To enable chronyd, enter:
   `sudo systemctl enable chronyd`
4. To restart chronyd, enter:
   `sudo systemctl restart chronyd`
5. To verify that the network time protocol service is running, enter:
   `sudo systemctl status chronyd.service`
6. To check that the server is synced to NTP, enter:
   `chronyc tracking`

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status
   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

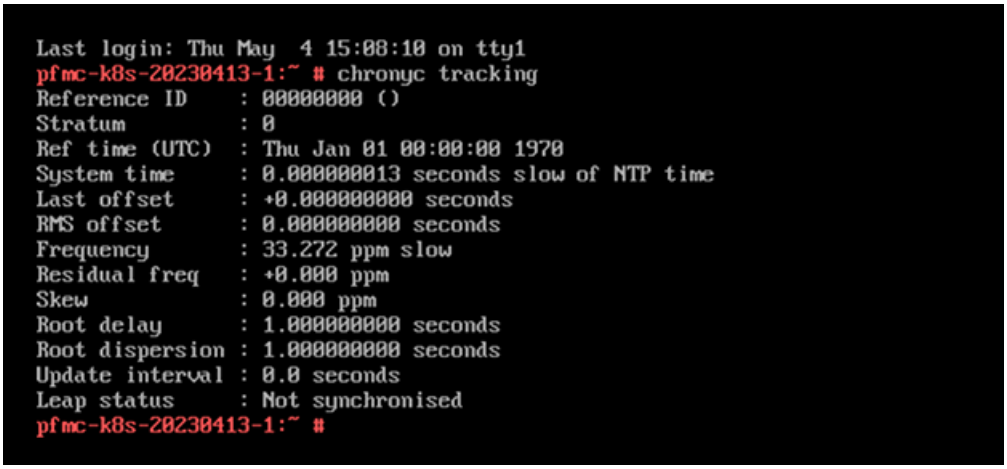
```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
System time : 0.000531569 seconds slow of NTP time
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

Unsuccessful chronyc tracking status

- `Stratum` is 0 (not synced).
- Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

The following image describes an unsuccessful chronyc tracking status.



7. Change the hostname, type the following: **`sudo hostnamectl set-hostname <mvm hostname>`**.

PowerFlex management platform cluster configuration parameters

The following table describes the configuration parameters for the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

The following diagram illustrates how PFMP_config.json file content is applied to PowerFlex management.

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

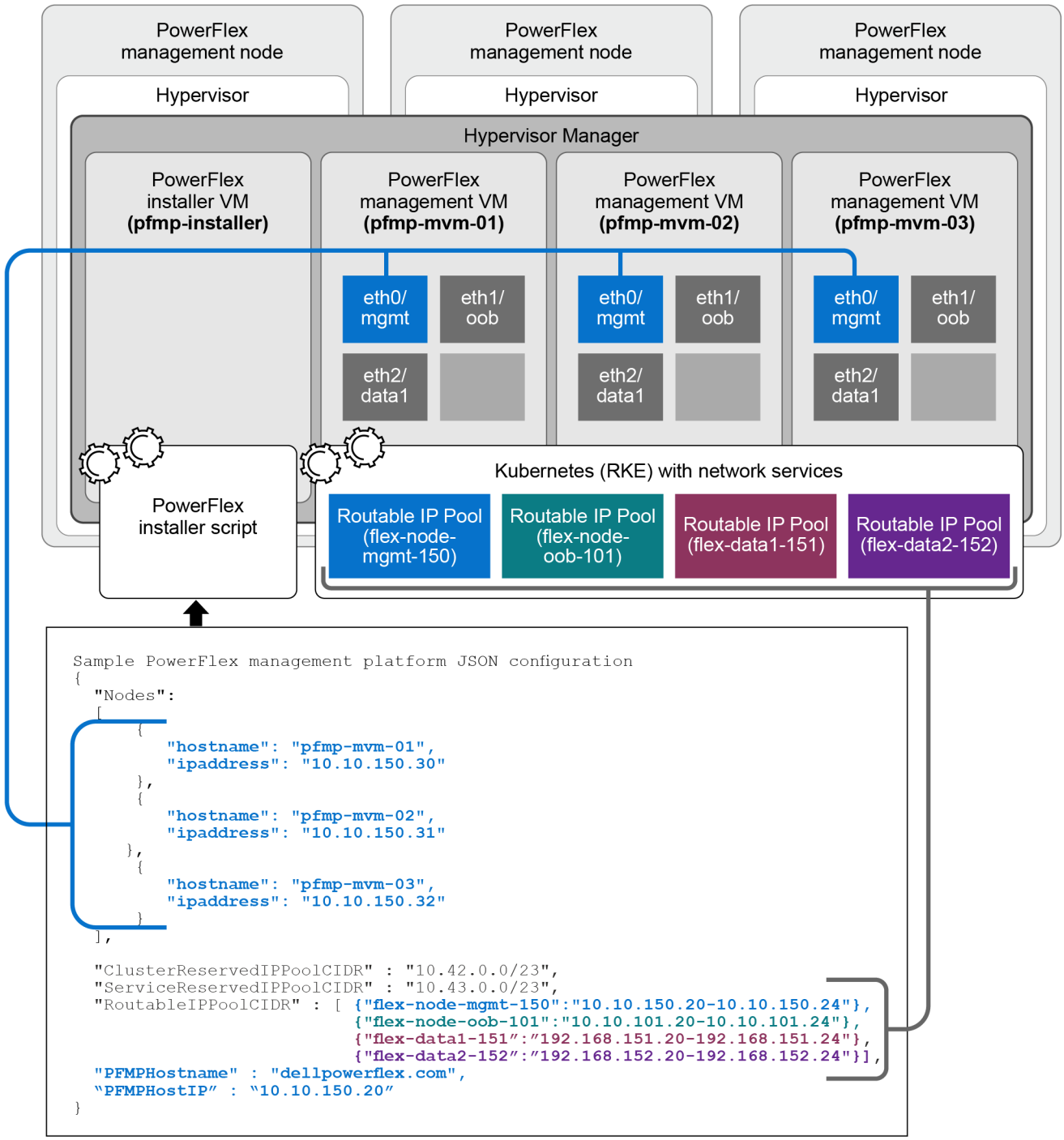


Figure 2. Relationship between the PFMP_config.JSON file and PowerFlex management

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 6. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Minimum of three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform

Table 6. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
ServiceReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and be lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform.

Steps

1. Using SSH, connect as a non-root user (**delladmin**) to the PowerFlex management platform installer VM and run the following command:

   ssh delladmin@<pfmp installer ip>

2. Access the `config` directory:
   `cd /opt/dell/pfmp/PFMP_Installer/config`
3. To configure the `PFMP_Config.json`, run **`sudo vi PFMP_Config.json`** and update the configuration parameters.
   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE:

- **`"RoutableIPPoolCIDR"`** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- PMFPHostIP has to be in the range of **`"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}`**.
- The **`"PFMPHostname" : "dellpowerflex.com"`** and **`"PFMPHostIP"`** are user defined and must have the appropriate DNS and PTR records created before installation.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Access the scripts folder in the working directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
5. Set up the installer:
   `sudo ./setup_installer.sh`

   The **`setup_installer.sh`** starts a docker container with the appropriate network settings.

6. Install the PowerFlex management platform:
   `sudo ./install_PFMP.sh`

   The **`install_PFMP.sh`** uses the docker installation to deploy PowerFlex Manager management cluster across the three virtual machines or hosts.

7. When prompted, do the following:
   - Enter **`No`** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **`Yes`**.
   - Enter the non-default SSH username for the nodes specified in `PFMP_Config.json`.
   - Enter **`Yes`** to indicate that the passwords are the same for all cluster nodes.
   - Enter the non-default SSH password for the nodes specified in `PFMP_Config.json`.

- Enter **No** to indicate that the nodes used for PowerFlex management platform are not integrated PowerFlex management (non-KVM) nodes.

NOTE: If you want to deploy a PowerFlex cluster using a CSV topology file, ensure that you use the single SSH key for all nodes.

View the progress of the installation in the following logs that are in the following directory:

- `/opt/dell/pfmp/PFMP_Installer/logs/`
- `/opt/dell/pfmp/atlantic/logs/`

NOTE: Installation may take more than two hours. When the deployment is complete, output similar to the following should appear:

```
UI can be accessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

- If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see *Related information*.
- For more information about troubleshooting, see *Related information*.

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
```

```
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platform cluster environment or uninstall the cluster.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
2. Run the following script to reset or uninstall the PowerFlex management platform cluster.
   `sudo ./reset_PFMP.sh`
   NOTE: If any troubleshooting issues are present, you can redeploy the cluster using *Deploy the PowerFlex management platform cluster*.

Remove the PowerFlex management platform installer VM

Use this procedure to remove the PowerFlex management platform installer VM.

Prerequisites

Before removing the PowerFlex management platform installer VM take a backup of the **`PFMP_Config.json`** file at `/opt/dell/pfmp/PFMP_Installer/config`. The **`PFMP_Config.json`** file is required when restoring PowerFlex Manager.

Steps

1. Log in to the VMware vCenter Server Appliance.
2. From the menu, click **Shortcuts** > **Hosts and Clusters**.
3. Right-click **PowerFlex management platform installer VM** > **Power** > **Power Off**.
4. Once powered off, right-click **Delete from Disk**.

Disable kdump on PowerFlex management cluster and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes.

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **`delladmin`**
2. Verify the service status, type the following: **`systemctl status kdump.service`**
3. Stop the kdump service, type the following: **`sudo systemctl stop kdump.service`**
4. Disable the kdump service, type the following: **`sudo systemctl disable kdump.service`**
5. Modify the GRUB configuration file, type the following:**`#vi /etc/default/grub`**

a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
b. Change the **crashkernel** parameter to **crashkernel=no**.

> **NOTE:** If the parameter is **crashkernel=<some_size>**, change it to **crashkernel=no**.

c. Save the file, press **ESC**+ **:wq**.

6. Update the GRUB configuration file, type the following: **sudo grub2-mkconfig -o /boot/grub2/grub.cfg**
7. Reboot the system, type the following: **sudo reboot**

Deploying the PowerFlex management platform into an integrated PowerFlex management (non-KVM) environment

Integrated PowerFlex management (non-KVM) deployment deploys the PowerFlex management platform onto the boot device of three PowerFlex storage nodes where the PowerFlex Manager and PowerFlex storage co-reside on the same node.

PowerFlex Manager is installed on a mirrored NVMe class boot device, such as one of the following:

- R660 BOSS-N1
- R650 BOSS-S2

The minimum NVMe disk size required is 960 GB.

After the operating system and PowerFlex Manager are installed, PowerFlex storage is deployed across the same management nodes through CSV.

Deployment workflow

The following table describes the main steps for deploying the PowerFlex management platform on the BOSS card in a integrated PowerFlex management (non-KVM) deployment.

Table 7. PowerFlex management platform integrated PowerFlex management (non-KVM) deployment workflow

Step	Action	Reference
1	Configure the BIOS settings and iDRAC (if applicable).	For Dell PowerFlex custom node, see the following guides available on the Dell Technologies support site. • *Dell PowerFlex Custom Node R660-R6625-R760-R7625 Hardware Configuration and Operating System Installation Guide for Linux* • *Dell PowerFlex Custom Node R650-R6525-R750 Hardware Configuration and Operating System Installation Guide for Linux*
2	Install the Linux operating system using vendor best practices. **NOTE:** When deploying the PowerFlex management platform in a integrated PowerFlex management (non-KVM) environment on a BOSS card, the maximum space should be allocated to the `/root` partition.	
3	Configure the network on the nodes and the PowerFlex management platform installer.	Configure the PowerFlex management platform installer node with the same parameters as the other nodes. When the PowerFlex management platform installer functionality is removed it will be used as a primary MDM.
4	Set up the YUM/Zypper repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP6*.
6	Deploy the PowerFlex management platform installer and run the PowerFlex management platform deployment.	See *Deploying the PowerFlex management platform*for more information.

Table 7. PowerFlex management platform integrated PowerFlex management (non-KVM) deployment workflow (continued)

Step	Action	Reference
7	Log in to PowerFlex Manager.	Verify that the PowerFlex Manager is deployed successfully.
8	Deploy PowerFlex with PowerFlex Manager on the integrated PowerFlex management (non-KVM) nodes using the CSV procedure.	See *Deploying a PowerFlex cluster using a CSV topology file* for more information.

Integrated PowerFlex management (non-KVM) deployment prerequisites

There are certain prerequisites that are required by PowerFlex Manager before you can deploy a Kubernetes cluster.

The following operating systems are supported:

- Red Hat Enterprise Linux 8.10, 9.6
- SUSE Linux Enterprise Server (SLES) SLES15 SP6

The following modules must be installed on the participating nodes before starting a integrated PowerFlex management (non-KVM) deployment.

The following list of modules must be installed on the participating nodes for SUSE Linux Enterprise Server before starting an integrated PowerFlex management (non-KVM) deployment. The modules are located on the install ISO.

- python3
- python3-pip
- sshpass
- java-11-openjdk-headless
- java-17-openjdk-headless
- jq
- haproxy
- keepalived
- lvm2
- skopeo
- apparmor-parser
- firewalld
- iptables
- libapr1

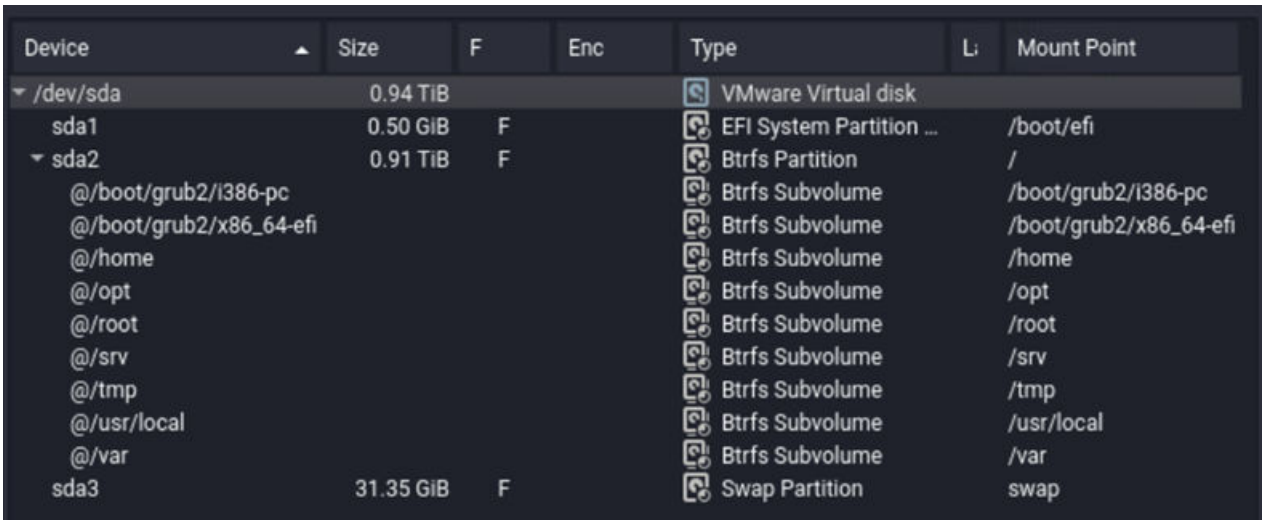
The following list of modules must be installed on the participating nodes for Red Hat Enterprise Linux before starting a integrated PowerFlex management (non-KVM) deployment. The modules are located on the install ISO.

- python3
- python3-pip
- httpd-tools
- python3-libselinux
- sshpass
- java-11-openjdk-headless
- java-17-openjdk-headless
- jq
- haproxy
- keepalived
- lvm2
- skopeo
- firewalld
- iptables

Complete the following:

1. All nodes must have their clocks synchronized with a valid time source.

2. The default route is configured before installing PowerFlex Manager.
3. When installing the operating system to the boot device, it is important to allocate maximum available space on the `/(root)` partition on all the four nodes.
   - In the following image all the remaining space (.91 TB) was allocated to the `/(root)` on partition sda1.



The following table describes the roles of the PowerFlex management platform deployment nodes.

Table 8. Roles of PowerFlex management platform deployment nodes

Number of nodes	Role	Recommended function
Node1	PFMP	Primary MDM
Node2	PFMP	Secondary MDM
Node3	PFMP	Secondary MDM
Node4	PFMP	Tiebreaker
Node5	PFMP	Tiebreaker
Node6	Installer	• Removes the installer functionality. • The installer can also be a temporary VM if it is a five-node deployment. The MDM roles will shift down to node2, node3, node 4, and node 5.

Deploy the PowerFlex management platform installer

Use the following procedure to download and prepare the PowerFlex management platform installer. The PowerFlex management platform installer is deployed temporarily on a physical server. Alternatively, you could deploy the PowerFlex management platform installer from a VM on a hypervisor external to the system that has network connectivity to the deployment nodes. After deployment, the PowerFlex management platform installer is removed or the installer VM can be deleted.

Prerequisites

The node that is used for the PowerFlex management platform installer will require access to meet one of the following requirements:

- To the Internet and attached to a SUSE account.
- Attached to a local SUSE RMT server at the customer site.
- Attached to a local zypper (SLES) or yum (Red Hat Enterprise Linux) repository at the customer site that contains the needed packages and their dependencies such as the installer ISO.

Steps

1. Log in to the installer node, which will become the primary MDM, and install docker.

   For SLES, run the following command:

   zypper install docker python3-docker-compose

   For Red Hat Enterprise Linux, run the following command:

   yum install podman-docker podman-plugins
2. Create and go to the `pfmp` directory with the following commands:

   sudo mkdir -p /opt/dell/pfmp

   cd /opt/dell/pfmp
3. From your desktop, download the PowerFlex installation file from the Dell Technologies support site.
   a. Click **Sign In** and log in with your account.
   b. Search for PowerFlex software and select it from the list.
   c. Go to **Drivers and Download** and select **PowerFlex Manager Bundle 5.0.x Build xxx**.
   d. Download the PowerFlex Manager bundle.
4. Copy the bundle to the `/opt/dell/pfmp` directory created on the first node.
5. Extract the package with the following command:

   sudo tar -xvf PowerFlex-Manager-bundle-5.0.x-Build-xxx.tgz

PowerFlex management platform cluster configuration parameters

The following table describes the configuration parameters for the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 9. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Minimum of three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
ServiceReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any

Table 9. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and be lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform cluster.

Steps

1. Go the `config` directory:
   `cd /opt/dell/pfmp/PFMP_Installer/config`
2. Open the `PFMP_Config.json` file for editing:
   `sudo vi PFMP_Config.json`
3. Update the configuration parameters in the `PFMP_Config.json` file according to the details for your deployment. Refer as necessary to the table in the *PowerFlex management platform cluster configuration parameters* topic in related information. .

   CAUTION: Modify the file carefully. Removing `[]` or `{}` causes the install script to fail.

   The configuration file varies according to the needs of the customer. For example, in the following example, there are one management interface and two data networks.

   NOTE: Ensure the hostname is the same as that defined on the physical node.

```
{
    "Nodes":
    [
      {
        "hostname": "mvm-245",
        "ipaddress": "10.234.89.245"
      },
      {
        "hostname": "mvm-246",
        "ipaddress": "10.234.89.246"
      },
```

```
        {
          "hostname": "mvm-247",
          "ipaddress": "10.234.89.247"
        }
      {
      "hostname": "mvm-248",
      "ipaddress": "10.234.89.248"
      }
      {
      "hostname": "mvm-249",
      "ipaddress": "10.234.89.249"
      }
      ],
      "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
      "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
      "RoutableIPPoolCIDR": [
      {
        "flex-node-mgmt-<vlanid>": "10.234.89.250 - 10.234.89.254"
      },
      {
        "data1-<vlanid>": "192.168.151.20- 192.168.151.24"
      },
      {
        "data2-<vlanid>": "192.168.152.20 - 192.168.152.24"
      }
    ],
      "PFMPHostname" : "powerflex_manager.fqdn.com",
      "PFMPHostIP" : "10.234.89.250"
}
```

NOTE:

- `RoutableIPPoolCIDR` requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- `PMFPHostIP` must be in the range of **`"RoutableIPPoolCIDR" : [{"mgmt":"10.234.89.250 - 10.234.89.254"}]`**.
- `PFMPHostname` and `PFMPHostIP` are user-defined and must have the appropriate DNS and PRT records created before the installation. `PFMPHostname` must be set to a hostname with a fully qualified domain name (FQDN) where the DNS resolves to the `PFMPHostIP`.
- The `PFMPHostname` value, *powerflex_manager.fqdn.com* is used here as an example.

Routed networks must be accessible through the interfaces on the hosts or virtual machines (VMs). If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the `RoutableIPPoolCIDR` section for that interface.

4. Access the scripts folder in the working directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
5. Ensure that the installer host has IPv4 forwarding enabled:
   a. On the installer host, run the following command:
      `sudo sysctl net.ipv4.ip_forward`
      Sample output: `net.ipv4.ip_forward = 1`
   b. If the command returns `0 net.ipv4.ip_forward = 0`, add **`net.ipv4.ip_forward = 1`** to the `/etc/sysctl.conf` file.
   c. Run the following command **`sudo sysctl -p`** to reload the configuration file.
6. Run the following command to set up the installer:
   `sudo ./setup_installer.sh`
   The **`setup_installer.sh`** installs and starts a docker container.
7. Run the following command to install the PowerFlex management platform.
   `sudo ./install_PFMP.sh`

8. Answer the following questions in the prompt::

```
Are ssh keys used for authentication connecting to the cluster nodes?
Answer= No (unless ssh keys are used)
Please enter the ssh username for the nodes specified in the PFMP_Config.json[root]
Answer: (your ssh username)
Are passwords the same for all the cluster nodes?
Answer: Yes
Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Answer: (your ssh password)
Are the nodes used for the PFMP cluster, co-res nodes?
Answer: Yes
```

NOTE: When deploying PowerFlex storage with a CSV file using SSH keys for the nodes, ensure that you use the same keys for all nodes.

Installation may take more than two hours. You can view the progress of the installation in the following logs, located in the directory:

- `/opt/dell/pfmp/PFMP_Installer/logs/`
- `/opt/dell/pfmp/atlantic/logs/`

When the deployment is complete, output similar to the following should appear:

```
UI can be acccessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

9. Use to access the PowerFlex Manager login screen.
10. In case of a failure or to check the progress, see the following log files:
    - PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
    - Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

    If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see *(Optional) Collect PowerFlex management platform support data*.

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
```

```
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platformcluster environment or uninstall the cluster.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
   If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see related information.
2. Run the following script to reset or uninstall a PowerFlex management platform cluster:
   `./reset_PFMP.sh`

Post deployment cleanup

After you deploy the PowerFlex management platform, perform the following steps to uninstall the PowerFlex management platform installer. The PowerFlex management platform installer is not longer needed after you deploy the PowerFlex management platform cluster.

Steps

1. Using SSH, connect to the node that the PowerFlex management platform installer is deployed on, and disable IPv4 forwarding:
   a. Open the `sysctl.conf` for editing:
      `sudo vi /etc/sysctl.conf`
   b. Modify the following line: `net.ipv4.ip forward = 1` to **`net.ipv4.ip forward = 0`**
   c. Run the following command to reload the configuration file:
      `sudo sysctl -p`
2. Delete the install directory with the following command:

```
rm -r /opt/dell/pfmp
```

3. Stop and remove the Docker/Podman container:

```
sudo systemctl disable docker

sudo systemctl stop docker

zypper remove docker python3-docker-compose
```

NOTE: Use the following command for Red Hat: **`yum remove podman-docker podman-plugins`**

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host.

About this task

NOTE: It is recommended to backup the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the temporary installer VM with the command **`sudo virsh console tempinstaller`**.
2. Press **Enter** to get the login prompt.
3. Log in as **`delladmin`** with the password **`delladmin`**.
4. Issue a command to perform a graceful shutdown of the VM: **`sudo virsh shutdown tempinstaller`**
5. Undefine the VM and remove the associated storage **`sudo virsh undefine tempinstaller --remove-all-storage`**
6. Verify the installer is no longer running / defined **`sudo virsh list --all`**

Disable kdump on PowerFlex management cluster (KVM) and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes (KVM).

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **`delladmin`**.
2. Verify the service status, type the following: **`systemctl status kdump.service`**
3. Stop the kdump service, type the following: **`sudo systemctl stop kdump.service`**
4. Disable the kdump service, type the following: **`sudo systemctl disable kdump.service`**
5. Modify the GRUB configuration file, type the following:**`#vi /etc/default/grub`**
   a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
   b. Change the **`crashkernel`** parameter to **`crashkernel=no`**.

      NOTE: If the parameter is **`crashkernel=<some_size>`**, change it to **`crashkernel=no`**.

   c. Save the file, press **ESC**+ **`:wq`**.
6. Update the GRUB configuration file, type the following: **`sudo grub2-mkconfig -o /boot/grub2/grub.cfg`**
7. Reboot the system, type the following: **`sudo reboot`**

Perform PowerFlex Manager initial configuration

After installation, use the following procedure to complete the initial configuration of PowerFlex Manager.

Steps

1. Open a web browser and enter **`https://<pfmphostname>`**, where **`<pfmphostname>`** is taken from the PFMP_config.json file.
2. Log in to PowerFlex Manager, see the *Dell PowerFlex Family 5.x Security Configuration Guide* for log in details.
3. At the first login, you are prompted to change the password.
4. On the **Welcome** page, read the instructions, and click **Next**.
5. If you want to enable SupportAssist, see related information, and click **Next**.
6. Select **I want to deploy a new instance of PowerFlex**.
7. Review the summary and click **Finish**.

Next steps

Deploy PowerFlex storage, for detailed procedures see related information.

Deploying the PowerFlex management platform into a Linux management environment

In a PowerFlex bring your own Linux management configuration, the PowerFlex management platform is installed on the Linux operating system directly.

Deployment workflow into a Linux management environment

The steps to deploy into a Linux management environment are as follows:

NOTE: The storage for PowerFlex management platform must be at least an SSD SAS level specification, it does not need to be shared storage, each node has its own partitions to use.

Table 10. Deployment workflow for a Linux management environment

Step	Action	Reference
1	Configure BIOS settings and the BMC controller (if applicable).	See the *Dell PowerFlex Custom Node R660-R6625-R760-R7625-R860 Hardware Configuration and Operating System Installation Guide for Linux* or *Dell Technologies PowerFlex Custom Node R650/R750/R6525 Hardware Configuration and Operating System Installation Guide for Linux* available at Dell Technologies support site.
2	Install the Linux operating system using the vendor's best practices.	When deploying the operating system reserve 600 GB for /var and 200 GB for /opt
3	Configure the network on the nodes.	N/A
4	Set up the yum \ zypper repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP6* available at the SUSE documentation website.
5	Deploy the PowerFlex management platform installer and run PowerFlex deployment.	See the *Deploying the PowerFlex management platform* for more information.
6	Log in to PowerFlex Manager and deploy a PowerFlex system.	To deploy PowerFlex, see *Deploy PowerFlex storage*.

Linux management deployment prerequisites

Ensure that the following prerequisites are met before deploying the PowerFlex management platform Kubernetes clusters:

Supported operating systems

- Red Hat Enterprise Linux 8.10, 9.6
- SUSE Linux Enterprise Server (SLES) 15 SP6

Required modules

The following modules must be installed on the participating nodes before starting the deployment:

Table 11. Deployment prerequisites for SUSE Linux Enterprise Server

Module	Location	Repository
python3	On ISO	N/A
python3-pip	On ISO	N/A
sshpass	SUSEConnect	PackageHub/15.x/x86_64
java-11-openjdk-headless	On ISO	N/A
java-11-openjdk-headless	On ISO	N/A
jq	On ISO	N/A
haproxy	SUSEConnect	sle-ha/15.x/x86_64
keepalived	SUSEConnect	sle-ha/15.x/x86_64
lvm2	On ISO	N/A
skopeo	SUSEConnect	sle-ha/15.x/x86_64
apparmor-parser	On ISO	N/A
libapr1	SUSEConnect	sle-module-basesystem/15.x/x86_64

Table 12. Deployment prerequisites for Red Hat Linux Enterprise

Module	Location	Repository
python3	On ISO	N/A
python3-pip	On Satellite	rhel-x-for-x86_64-appstream-rpms
httpd-tools	On Satellite	rhel-x-for-x86_64-appstream-rpms
python3-libselinux	On ISO	N/A
sshpass	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-11-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-17-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms
jq	On Satellite	rhel-x-for-x86_64-appstream-rpms
haproxy	On Satellite	rhel-x-for-x86_64-appstream-rpms
keepalived	On Satellite	rhel-x-for-x86_64-appstream-rpms
lvm2	On ISO	N/A
skopeo	On Satellite	rhel-x-for-x86_64-appstream-rpms

Volumes and mapping commands

Dedicate all unused storage to the `/(root)` partition.

Additional prerequisites

- All nodes must have their clocks synchronized with a valid time source.
- Ensure that the default route is configured before installing the PowerFlex management platform.

Deployment design for Linux management

The following details best practices in relation to component layout:

NOTE: The following steps are also applicable for a Linux VM running on top of KVM. The KVM configuration is not in this guide, and customers can manage the setup:

- Number of nodes: 3
- Node 1: Kubernetes for the PowerFlex management platform
- Node 2: Kubernetes for the PowerFlex management platform
- Node 3: Kubernetes for the PowerFlex management platform

Configuring Linux management nodes

Use the procedures in this section to configure Linux management nodes.

Setting up the PowerFlex management platform

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

Deploy the PowerFlex management platform installer

Use the following procedure to download and prepare the PowerFlex management platform installer. The PowerFlex management platform installer is deployed temporarily on what will become the primary PowerFlex MDM. Alternatively, you could deploy the PowerFlex management platform installer from a VM on a hypervisor external to the system that has network connectivity to the deployment nodes. After deployment, the PowerFlex management platform installer is removed or the installer VM can be deleted.

Prerequisites

The node that is used for the PowerFlex management platform installer will require access to meet one of the following requirements:

- To the Internet and attached to a SUSE account.
- Attached to a local SUSE RMT server at the customer site.
- Attached to a local zypper (SLES) or yum (Red Hat Enterprise Linux) repository at the customer site that contains the needed packages and their dependencies such as the installer ISO.

Steps

1. Log in to the installer node, which will become the primary MDM, and install docker.

   For SLES, run the following command:

   zypper install docker python3-docker-compose

   For Red Hat Enterprise Linux, run the following command:

   yum install podman-docker podman-plugins
2. Create and go to the `pfmp` directory with the following commands:

   sudo mkdir -p /opt/dell/pfmp

   cd /opt/dell/pfmp
3. From your desktop, download the PowerFlex installation file from the Dell Technologies support site.
   a. Click **Sign In** and log in with your account.
   b. Search for PowerFlex software and select it from the list.
   c. Go to **Drivers and Download** and select **PowerFlex Manager Bundle 5.0.x Build xxx**.
   d. Download the PowerFlex Manager bundle.
4. Copy the bundle to the `/opt/dell/pfmp` directory created on the first node.
5. Extract the package with the following command:

   sudo tar -xvf PowerFlex-Manager-bundle-5.0.x-Build-xxx.tgz

PowerFlex management platform cluster configuration parameters

The following table describes the configuration parameters for the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

NOTE: For PowerFlex software, including the `oob` VLAN is optional.

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 13. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Minimum of three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
ServiceReservedIPPoolCIDR	IP address	Minimum / 23 network The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any

Table 13. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		existing subnets in the datacenter. Users could use the same subnet for multiple PowerFlex management platform deployments. The default subnets 10.43.x.x and 10.42.x.x, as listed in the JSON file, can be used. If these defaults conflict with the your environment, they can be modified accordingly.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and be lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Deploy the PowerFlex management platform cluster

Use the following procedure to deploy the PowerFlex management platform cluster.

Steps

1. Go the `config` directory:
   `cd /opt/dell/pfmp/PFMP_Installer/config`
2. Open the `PFMP_Config.json` file for editing:
   `sudo vi PFMP_Config.json`
3. Update the configuration parameters in the `PFMP_Config.json` file according to the details for your deployment. Refer as necessary to the table in *PowerFlex management platform cluster configuration parameters*.

   CAUTION: Modify the file carefully. Removing `[]` or `{}` causes the install script to fail.

   The configuration file varies according to the needs of the customer. In the following example, there are one management interface and two data networks.

   NOTE: Ensure the hostname is the same as that defined on the physical node.

```
{
    "Nodes":
    [
      {
        "hostname": "mvm-245",
        "ipaddress": "10.234.89.245"
      },
      {
        "hostname": "mvm-246",
        "ipaddress": "10.234.89.246"
      },
      {
```

```
        "hostname": "mvm-247",
        "ipaddress": "10.234.89.247"
      }
    ],
    "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
    "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
    "RoutableIPPoolCIDR": [
    {
      "flex-node-mgmt-<vlanid>": "10.234.89.250 - 10.234.89.254"
    },
    {
      "data1-<vlanid>": "192.168.151.20- 192.168.151.24"
    },
    {
      "data2-<vlanid>": "192.168.152.20 - 192.168.152.24"
    }
  ],
    "PFMPHostname" : "powerflex_manager.fqdn.com",
    "PFMPHostIP" : "10.234.89.250"
}
```

NOTE:

- `RoutableIPPoolCIDR` requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address.
- `PMFPHostIP` must be in the range of **`"RoutableIPPoolCIDR" : [{"mgmt":"10.234.89.250 - 10.234.89.254"}]`**.
- `PFMPHostname` and `PFMPHostIP` are user-defined and must have the appropriate DNS and PRT records created before the installation. `PFMPHostname` must be set to a hostname with a fully qualified domain name (FQDN) where the DNS resolves to the `PFMPHostIP`.
- The `PFMPHostname` value, *powerflex_manager.fqdn.com* is used here as an example.

Routed networks must be accessible through the interfaces on the hosts or virtual machines (VMs). If PowerFlex Manager requires access to non-routed networks, add the required interfaces to the host or VM and add an additional line to the `RoutableIPPoolCIDR` section for that interface.

4. Access the scripts folder in the working directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
5. Ensure that the installer host has IPv4 forwarding enabled:
   a. On the installer host, run the following command:
      `sudo sysctl net.ipv4.ip_forward`
      Sample output: `net.ipv4.ip_forward = 1`
   b. If the command returns `0 net.ipv4.ip_forward = 0`, add **`net.ipv4.ip_forward = 1`** to the `/etc/sysctl.conf` file.
   c. Run the following command **`sudo sysctl -p`** to reload the configuration file.
6. Run the following command to set up the installer:
   `sudo ./setup_installer.sh`
   The **`setup_installer.sh`** installs and starts a docker container.
7. Run the following command to install the PowerFlex management platform.
   `sudo ./install_PFMP.sh`
8. Answer the following questions in the prompt::

```
Are ssh keys used for authentication connecting to the cluster nodes?
Answer= No (unless ssh keys are used)
Please enter the ssh username for the nodes specified in the PFMP_Config.json[root]
Answer: (your ssh username)
Are passwords the same for all the cluster nodes?
Answer: Yes
Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Answer: (your ssh password)
```

```
Are the nodes used for the PFMP cluster, co-res nodes?
Answer: No
```

NOTE: When deploying PowerFlex storage with a CSV file using SSH keys for the nodes, ensure that you use the same keys for all nodes.

Installation may take more than two hours. You can view the progress of the installation in the following logs, located in the directory:

- `/opt/dell/pfmp/PFMP_Installer/logs/`
- `/opt/dell/pfmp/atlantic/logs/`

When the deployment is complete, output similar to the following should appear:

```
UI can be acccessed at: powerflex_manager.fqdn.com which needs to be resolved to
<PFMPHostIP>
```

9. Use **`powerflex_manager.fqdn.com`** to access the PowerFlex Manager login screen.
10. In case of a failure or to check the progress, see the following log files:
    - PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
    - Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

    If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see *(Optional) Collect PowerFlex management platform support data*.

(Optional) Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform installation issues.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site.
   - Get the utility file from the PowerFlex management platform installer: Go to the `PFMP_Installer/scripts` directory.
   - Download the data collection utility from the Dell Technologies support site: Search for **PFMP log collection Script** file on the Dell Technologies support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as root user.

   For example, the utility executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, use the following steps to reset the PowerFlex management platformcluster environment or uninstall the cluster.

Steps

1. Change the directory to the PowerFlex management platform installer scripts directory:
   `cd /opt/dell/pfmp/PFMP_Installer/scripts`
   If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see related information.
2. Run the following script to reset or uninstall a PowerFlex management platform cluster:
   `./reset_PFMP.sh`

Post deployment cleanup

After you deploy the PowerFlex management platform, perform the following steps to uninstall the PowerFlex management platform installer. The PowerFlex management platform installer is no longer needed after you deploy the PowerFlex management platform cluster.

Steps

1. Using SSH, connect to the node that the PowerFlex management platform installer is deployed on, and disable IPv4 forwarding:
   a. Open the `sysctl.conf` for editing:
      `sudo vi /etc/sysctl.conf`
   b. Modify the following line: `net.ipv4.ip forward = 1` to **`net.ipv4.ip forward = 0`**
   c. Run the following command to reload the configuration file:
      `sudo sysctl -p`
2. Delete the install directory with the following command:
   `rm -r /opt/dell/pfmp`
3. Stop and remove the docker/podman container:
   `sudo systemctl disable docker`

   `sudo systemctl stop docker`

   `zypper remove docker python3-docker-compose`

   Use the following command for Red Hat:

```
yum remove podman-docker podman-plugins
```

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host.

About this task

NOTE: It is recommended to backup the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the temporary installer VM with the command **`sudo virsh console tempinstaller`**.
2. Press **Enter** to get the login prompt.
3. Log in as **`delladmin`** with the password **`delladmin`**.
4. Issue a command to perform a graceful shutdown of the VM: **`sudo virsh shutdown tempinstaller`**
5. Undefine the VM and remove the associated storage **`sudo virsh undefine tempinstaller --remove-all-storage`**
6. Verify the installer is no longer running / defined **`sudo virsh list --all`**

Disable kdump on PowerFlex management cluster (KVM) and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes (KVM).

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **`delladmin`**.
2. Verify the service status, type the following: **`systemctl status kdump.service`**
3. Stop the kdump service, type the following: **`sudo systemctl stop kdump.service`**
4. Disable the kdump service, type the following: **`sudo systemctl disable kdump.service`**
5. Modify the GRUB configuration file, type the following:**`#vi /etc/default/grub`**
   a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
   b. Change the **`crashkernel`** parameter to **`crashkernel=no`**.

      NOTE: If the parameter is **`crashkernel=<some_size>`**, change it to **`crashkernel=no`**.

   c. Save the file, press **ESC**+ **`:wq`**.
6. Update the GRUB configuration file, type the following: **`sudo grub2-mkconfig -o /boot/grub2/grub.cfg`**
7. Reboot the system, type the following: **`sudo reboot`**

Perform PowerFlex Manager initial configuration

After installation, use the following procedure to complete the initial configuration of PowerFlex Manager.

Steps

1. Open a web browser and enter **`https://<pfmphostname>`**, where **`<pfmphostname>`** is taken from the PFMP_config.json file.
2. Log in to PowerFlex Manager, see the *Dell PowerFlex Family 5.x Security Configuration Guide* for log in details.

3. At the first login, you are prompted to change the password.
4. On the **Welcome** page, read the instructions, and click **Next**.
5. If you want to enable SupportAssist, see related information, and click **Next**.
6. Select **I want to deploy a new instance of PowerFlex**.
7. Review the summary and click **Finish**.

Next steps

Deploy PowerFlex storage, for detailed procedures see related information.

Deploying the PowerFlex management platform into a PowerFlex management cluster (KVM) environment

Use this section to deploy the Powerflex management cluster on a KVM environment.

Deploy and configure production MVM images

Use this procedure to deploy and configure production MVM images.

Steps

1. Log in to the first host as **delladmin**.
2. Download the QCOW image from the Dell Support site and copy the `pfmp-k8s-xxx-<date>.qcow2` image to all the three nodes on `/var/lib/libvirt/images/mvm-name`.
3. Deploy the MVM images using the following commands:

```
$ sudo virt-install --name mvm-name1 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole

$ sudo virt-install --name mvm-name2 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole

$ sudo virt-install --name mvm-name3 --ram 32768 --disk /var/lib/libvirt/
images/mvm-name --import --vcpus=14 --network bridge=bridge-<mgmt-VLAN>,model=virtio
--network bridge=bridge-<oob VLAN>,model=virtio --network bridge=bridge-<data1
VLAN>,model=virtio --network bridge=bridge-<data2 VLAN>,model=virtio --osinfo sle15 --
noautoconsole
```

4. Start the VMs:

   This should be run on all the 3 nodes where MVM is deployed:

```
$ sudo virsh start <mvm-name1>
$ sudo virsh start <mvm-name2>
$ sudo virsh start <mvm-name3>
```

Configure production MVMs

Use this procedure to configure the MVMs.

Steps

1. On the individual hosts, run the following command: **`sudo virsh console <mvm-name>`**.

2. Press **Enter** to get the login prompt.
3. Log in as **delladmin** with the password **delladmin**.
4. Change the hostname, type **sudo hostnamectl set-hostname <mvm hostname>**.
5. Configure the flex-stor-mgmt-<vlanid> eth0 interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=1500
```

   b. Configure the default route and run the following command: **sudo vi /etc/sysconfig/network/routes. default <gateway ip> - <interface>**.

   For example: **default 10.10.150.1 - eth0**.

   c. Configure the DNS search and DNS servers. Run the following command: **sudo vi /etc/sysconfig/network/config** and modify the following:

   NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

   NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

   For example:

   NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

   NETCONFIG_DNS_STATIC_SERVERS=" 10.10.105.240 10.10.105.241"

6. Configure the eth1 interface with flex-oob-mgmt-<vlanid> (if OOB is active) or flex-data1-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth1**

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu> # 1500 for OOB or 9000 for data network
```

   For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

7. Configure the eth2 interface with flex-data1-<vlanid> (if OOB is active) or flex-data2-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth2**

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
```

```
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

8. Configure the eth3 interface with flex-data2-<vlanid> (if OOB is active) or flex-data3-<vlanid> interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth3**

```
DEVICE=eth3
NAME=eth3
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth3
NAME=eth3
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

9. Configure the eth4 interface with flex-data3-<vlanid> (if OOB is active) or flex-data4-<vlanid>interface as follows:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth4**

```
DEVICE=eth4
NAME=eth4
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth4
NAME=eth4
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

10. Configure the eth5 interface with flex-data4-<vlanid> (if OOB is active) interface as follows:
    a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth5**

```
DEVICE=eth5
NAME=eth5
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
MTU=<mtu>
```

For example:

```
DEVICE=eth5
NAME=eth5
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.150.10
NETMASK=255.255.255.0
MTU=9000
```

11. To restart the IP addresses and networks, type **sudo systemctl restart network**.
12. Exit the console, close the window.
13. Repeat the above for the other production MVMs on the hosts theya re running on, gathered from step 1.

Configure the NTP service

Use this procdure to configure the NTP service.

Steps

1. SSH to the first MVM as **delladmin**.
2. To configure the NTP server, type **sudo vi /etc/chrony.conf** and add the following: **server <ntp ip address> iburst**.

   For example: **server 10.10.105.240 iburst**
3. To enable chronyd, type **sudo systemctl enable chronyd**.
4. To reboot, type **sudo reboot**.
5. Log in as **delladmin**
6. To check the server is synced to NTP, type **chronyc tracking**.

   NOTE: This may take a few minutes to sync. Examples of chronyc tracking status are as follows:

Table 14. Chronyc tracking status

Successful chronyc tracking status	Unsuccessful chronyc tracking status

A configured NTP server should be listed under `Reference ID`, `Stratum` should be a value greater than 0 and `System Time` should report much less than 1 second time difference:
`node1:~ # chronyc tracking`
`Reference ID : 64400012 (0.pool.ntp.org)`
`Stratum : 4`
`Ref time (UTC) : Mon Mar 07 17:15:39 2022`
`System time : 0.000001647 seconds slow of NTP time`
`Last offset : -0.000006101 seconds`
`RMS offset : 0.000020258 seconds`
`Frequency : 33.247 ppm slow`
`Residual freq : -0.000 ppm`
`Skew : 0.016 ppm`
`Root delay : 0.040418178 seconds`
`Root dispersion : 0.033837218 seconds`
`Update interval : 1024.6 seconds`
`Leap status : Normal`
	Stratum 0 means not synced and Leap status says `Not synchronized`:
	`node1:~ # chronyc tracking`
	`Reference ID : 00000000 ()`
	`Stratum : 0`
	`Ref time (UTC) : Thu Jan 01 00:00:00 1970`
	`System time : 0.000000000 seconds fast of NTP time`
	`Last offset : +0.000000000 seconds`
	`RMS offset : 0.000000000 seconds`
	`Frequency : 0.000 ppm slow`
	`Residual freq : +0.000 ppm`
	`Skew : 0.000 ppm`
	`Root delay : 1.000000000 seconds`
	`Root dispersion : 1.000000000 seconds`
	`Update interval : 0.0 seconds`
	`Leap status : Not synchronized`

7. Repeat these steps for all MVMs.

Deploy the temporary installer VM

A temporary installer VM is required to install the second PowerFlex Manager instance.

About this task

The VM is placed on local storage of one of the existing hosts and removed once the installation is complete.

Prerequisites

Ensure the QCOW2 image for the PowerFlex management platform installer VM must be accessible to the system.

Steps

1. SSH to one of the PowerFlex Manager controller hosts.
2. Download and copy the QCOW2 image for the installer to `/var/lib/libvirt/images/`.
3. Deploy the virtual machine, type:

```
$ sudo virt-install --name tempinstaller --ram 8192 --vcpus=8  --disk /var/lib/
libvirt/images/<img name>.qcow2   --network bridge=bridge-<mgmt-VLAN>,model=virtio --
osinfo sle15 --import --noautoconsole
```

Configure temporary installer

Steps

1. Connect to the temporary installer VM with the command **`sudo virsh console tempinstaller`**.
2. Press **Enter** to get the login prompt.
3. Log in as **`delladmin`** with the password **`delladmin`**.
4. Configure the flex-stor-mgmt-<vlanid> eth0 interface:
   a. Type: **`sudo vi /etc/sysconfig/network/ifcfg-eth0`**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

   b. Configure the default route. Type:

```
sudo vi /etc/sysconfig/network/routes
default<gateway ip> - <interface>
```

   For example: **`default 10.10.105.1 - eth0`**.

   c. Configure DNS search and DNS servers. Type: **`sudo vi /etc/sysconfig/network/config`** and modify the following:

   `NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"`

   `NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"`

For example:

```
NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"
NETCONFIG_DNS_STATIC_SERVERS=" 10.10.105.240 10.10.105.241"
```

5. To restart the IP addresses and network, type: **sudo systemctl restart network**.

Deploy the PowerFlex management platform cluster

Use this task to complete the steps needed to configure the `PFMP_Config.json` file which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following table describes the `PFMP_Config.json` and its configuration parameters.

NOTE: When configuring the RoutableIPPoolCIDR parameter in `PFMP_Config.json`, the subset of the IP addresses must be contiguous. It is five contiguous IP addresses for each flex-node-mgmt-<vlanid>, flex-oob-mgmt-<vlanid>, flex-data1-<vlanid>, and flex-data2-<vlanid>.

Table 15. PFMP_config.JSON and configuration parameters

Key	Value	Description
Nodes (minimum three VMs / nodes)	hostname IP address	The IP addresses of the nodes on which the PowerFlex management platform cluster will be deployed, along with the hostnames that need to be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR (minimum / 23 network)	IP address	The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Every PowerFlex management platform instance must use unique subnets. For example, PFMP-PROD should use different subnets than the PFMP-MGMT instance as well as all subsequent PowerFlex management platform instances.
ServiceReservedIPPoolCIDR (minimum / 23 network)	IP address	The minimum recommended subnet size is /23. Ensure that the private subnet does not conflict with any existing subnets in the datacenter. Every PowerFlex management platform instance must use unique subnets. For example, PFMP-PROD should use different subnets than the PFMP-MGMT instance as well as all subsequent PowerFlex management platform instances.
RoutableIPPoolCIDR	flex-stor-mgmt-150 flex-oob-mgmt-<vlanid> flex-data1-<vlanid>	IP requirements (number of IP addresses): Five contiguous IP addresses • flex-stor-mgmt-150 (5) • flex-oob-mgmt-<vlanid> (5) • flex-data1-<vlanid> (5)

Table 15. PFMP_config.JSON and configuration parameters (continued)

Key	Value	Description
	flex-data2-<vlanid> flex-data3-<vlanid> (if required) flex-data4-<vlanid> (if required)	• flex-data2-<vlanid> (5) • flex-data3-<vlanid> (5) (if required) • flex-data4-<vlanid> (5) (if required) IP addresses are for the following services: • Ingress (web server) IP address • HTTP IP address • SNMP IP address (receiving SNMP traps) • SupportAssist remote • Additional IP for HTTP services (will be used only during upgrade)
PFMPHostname	FQDN	This is the FQDN or hostname which a user can connect to the PowerFlex management platform UI using a browser. The FQDN/hostname needs to be resolvable through DNS or through a host file (where the browser is running).
PFMPHostIP	IP address	This parameter specifics the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer, run the following command: **ssh delladmin@<pfmp installer ip>**.
2. To navigate to the config directory, run the following command: **cd /opt/dell/pfmp/PFMP_Installer/config**
3. To configure the `PFMP_Config.json`, run the following command: **sudo vi PFMP_Config.json** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.150.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.150.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.150.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-stor-mgmt-150":"10.10.150.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.150.20"
}
```

NOTE: PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" : {"flex-stor-mgmt-<vlanid>":"10.10.150.20-10.10.150.24"}**.

NOTE: If the customer is planning for four data networks, add flex-data3-<vlanid> and flex-data4-<vlanid> also in PFMP_Config.json file.

4. To set up the installer, run the following command: **sudo ./setup_installer.sh**.
5. Navigate to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**

   NOTE: If the PowerFlex management platform has already been deployed on the management VMs, and you need to redeploy the PowerFlex management platform, the previous installation must be removed and cleaned from the VMs, type the following command: **sudo ./reset_installer.sh**.

   The reset_installer.sh script will provide the same prompts as the setup_installer.sh script, but removes the deployed cluster.

6. To install PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh** (enter sudo user and password for target nodes when prompted, when prompted, answer the questions provided.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`. The default user is delladmin.
   - Enter **Yes** to indicate that passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not co-resident nodes.
8. Once the installer is complete, use **powerflex_manager.fqdn.com** to log in to PowerFlex Manager using username and password.

   If a failure has occurred with PowerFlex management platform installer, the relevant errors are located at: `/opt/dell/pfmp/atlantic/logs/bedrock.log`. For more information on troubleshooting, see *Troubleshooting a PowerFlex management platform deployment*.

Remove temporary installer VM

Once installation is complete, the temporary installer VM may be removed from the host.

About this task

NOTE: It is recommended to backup the `PFMP_config.json` file used to the jump server to deploy the production PowerFlex Manager instance.

Steps

1. Connect to the temporary installer VM with the command **sudo virsh console tempinstaller**.
2. Press **Enter** to get the login prompt.
3. Log in as **delladmin** with the password **delladmin**.
4. Issue a command to perform a graceful shutdown of the VM: **sudo virsh shutdown tempinstaller**
5. Undefine the VM and remove the associated storage **sudo virsh undefine tempinstaller --remove-all-storage**
6. Verify the installer is no longer running / defined **sudo virsh list --all**

Disable kdump on PowerFlex management cluster (KVM) and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes (KVM).

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **delladmin**.
2. Verify the service status, type the following: **systemctl status kdump.service**
3. Stop the kdump service, type the following: **sudo systemctl stop kdump.service**
4. Disable the kdump service, type the following: **sudo systemctl disable kdump.service**
5. Modify the GRUB configuration file, type the following:**#vi /etc/default/grub**
   a. Locate the line starting with GRUB_CMDLINE_LINUX_DEFAULT.
   b. Change the **crashkernel** parameter to **crashkernel=no**.

      NOTE: If the parameter is **crashkernel=<some_size>**, change it to **crashkernel=no**.

   c. Save the file, press **ESC**+ **:wq**.
6. Update the GRUB configuration file, type the following: **sudo grub2-mkconfig -o /boot/grub2/grub.cfg**
7. Reboot the system, type the following: **sudo reboot**

5

Troubleshooting a PowerFlex management platform deployment

This troubleshooting section contains verification steps and methods that can be used to identify and address misconfigurations to ensure a successful PowerFlex management platform deployment.

The PowerFlex management platform runs on Kubernetes across three virtual machines that reside on controller nodes or a customer-provided virtual environment. The PowerFlex management platform deployment process can fail for several reasons. To ensure the expected state of the PowerFlex installer and management virtual machines (MVMs) before initiating the deployment.

Verifying the network time protocol setup

Use this section to verify the network time protocol (NTP) servers are synchronized before the install is initiated.

Time synchronization across all hosts and virtual machines is required. Functioning NTP servers are required for deployment. All PowerFlex nodes and virtual machines must point to a functioning NTP server and time must be verified as synchronized before the install is attempted.

Verify the NTP setup on a VMware ESXi-based PowerFlex node

Use the following procedure to verify the NTP is set up properly on a VMware ESXi-based PowerFlex node.

Steps

1. In the VMware vSphere Client home page, go to **Home** > **Hosts and Clusters**.
2. Select a host.
3. On the **Configure** tab, select **System** > **Time Configuration**.
4. Click **Test services**.
5. Verify that the check passes and time are accurate.

Verify the NTP setup on a Linux-based PowerFlex node

Use the following procedure to verify the NTP setup on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. Type the following: **`chronyc tracking`**.

   Output should have a valid reference ID and time should be accurate. The following gives an example output:

```
pfmp-mvm-01:~ # chronyc tracking
Reference ID    : 0AEA7C8B (dns.example.com)
Stratum         : 4
Ref time (UTC)  : Tue Jan 30 16:09:53 2024
System time     : 0.000041504 seconds slow of NTP time
Last offset     : -0.000069052 seconds
RMS offset      : 0.000188316 seconds
Frequency       : 16.336 ppm slow
Residual freq   : -0.011 ppm
Skew            : 0.077 ppm
Root delay      : 0.001050830 secondsRoot dispersion : 0.005389712 seconds
Update interval : 1034.1 seconds
```

```
Leap status     : Normal
pfmp-mvm-01:~ #
```

If chronyc tracking returns invalid data, see the NTP setup in *Configure the NTP service*.

Verifying the DNS

PowerFlex uses mTLS and SSL certificates across all services, hosts, and virtual machines. To get proper host resolution valid DNS A-records and PTR records are required. Use this section to verify the DNS is functioning as required.

Verify DNS on a VMware ESXi-based PowerFlex node

Use the following procedure to verify DNS on a VMware ESXi-based PowerFlex node.

Steps

1. SSH to the VMware ESXi-based PowerFlex node.
2. Type the following: **`nslookup <FQDN>`**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup pfmp-r650-01.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-r650-01.example.com
Address: 10.10.105.3

[root@pfmp-r650-01:~]
```

3. Type the following: **`nslookup <IP address>`**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup 10.10.105.3
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-r650-01.example.com

[root@pfmp-r650-01:~]
```

Verify DNS on a Linux-based PowerFlex node

Use the procedure to modify or verify DNS on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. To update or modify the DNS entries:
   a. Go the file path, run the following command: `sudo vi /etc/sysconfig/network/config`
   b. Add or modify the line:
      - **`NETCONFIG_DNS_STATIC_SEARCHLIST="xxx.com"`**, where **`xxx.com`** is the domain name.
      - **`NETCONFIG_DNS_STATIC_SERVERS="x.x.x.x y.y.y.y"`**, where **`x.x.x.x`** and **`y.y.y.y`** with your DNS server IP address.
   c. Save the file, press **Esc**+ **:wq**.
   d. Refresh and update the configuration, run the following command: **`sudo netconfig update -f`**.

Verifying the network

Use this section to verify the network. The PowerFlex management platform requires both internal and external network communication. See the *Dell PowerFlex Appliance and PowerFlex Rack with PowerFlex Cabling and Connectivity Guide* for more information.

Verify each of the three PowerFlex management virtual machines has the required interfaces:

- flex-node-mgmt-<vlanid>
- flex-oob-mgmt-<vlanid> (if required)
- flex-data1-<vlanid>
- flex-data2-<vlanid>
- flex-data3-<vlanid> (if required)
- flex-data4-<vlanid> (if required)

Verify the management virtual machine network communication

Use the following procedure to verify the management virtual machine network communication using a ping command.

Steps

1. SSH to each PowerFlex management virtual machine node.
2. For flex-node-mgmt-<vlanid> and flex-oob-mgmt-<vlanid>, type the following: **`ping <peer IP address>`**.
3. For flex-data1-<vlanid>, flex-data2-<vlanid>, flex-data3-<vlanid> (if required), and flex-data4-<vlanid> (if required), type the following: **`ping <peer IP address>`**.

   Using the following examples, verify all interfaces within the same subnet are reachable:

```
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-02 mgmt/eth0
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0

pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-02 oob/eth1
pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-03 oob/eth1

pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-02 data1/eth2
pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-03 data1/eth2

pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-02 data2/eth3
pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-03 data2/eth3

pfmp-mvm-02 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0
pfmp-mvm-02 oob/eth1 <-> pfmp-mvm-03 oob/eth1

pfmp-mvm-02 data1/eth2 <-> pfmp-mvm-03 data1/eth2
pfmp-mvm-02 data2/eth3 <-> pfmp-mvm-03 data2/eth3
```

   Example output:

```
pfmp-mvm-01:~ # ping 10.10.105.31
PING pfmp-mvm-02 (10.10.105.31) 56(84) bytes of data.
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=3 ttl=64 time=0.086 ms

--- from pfmp-mvm-02 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verify routing

Use the following procedure to verify the routing of networks.

About this task

Verify the routing for the following networks:

- flex-node-mgmt-<vlanid> route to flex-stor-mgmt-<vlanid>
- flex-node-mgmt-<vlanid> route to pfmc-stor-mgmt<vlanid>

Steps

1. SSH to each PowerFlex management virtual machine on flex-node-mgmt-<vlanid>.
2. Type the following: **ping <pfmc-stor-mgmt-<vlandid> target ip address>**.
3. Type the following: **ping <flex-stor-mgmt-<vlanid> target ip address>**.
   Example output:

```
pfmp-mvm-01:~ # ping 10.10.150.10
PING 10.10.150.10 (10.10.150.10) 56(84) bytes of data.
64 bytes from 10.10.150.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.150.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.150.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.150.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~


pfmp-mvm-01:~ # ping 10.10.140.10
PING 10.10.140.10 (10.10.140.10) 56(84) bytes of data.
64 bytes from 10.10.140.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.140.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.140.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.140.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verifying the PFMP_config.json file

Use this section to verify the PFMP_config.json file.

The PowerFlex management platform deployment process requires a configuration file to be filled out with customer network information. The `install_pfmp.sh` script uses the data in PFMP_config.json to deploy all the required packages, install Kubernetes, and deploy all the pods that are required to run PowerFlex Manager.

The template file is found on the installer VM under: `/opt/dell/pfmp/PFMP_Installer/config/PFMP_Config.json`.

Verify the PFMP_config.json key input restrictions

Use this procedure to verify the key input restrictions for the PFMP_config.json file.

Steps

Verify the following PFMP_config.json key input restrictions:

- Ensure that all hostnames and network names are in lowercase.
- Management virtual machine hostname values must NOT be fully qualified (FQDN).
- ClusterReserved and ServiceReserved Pools are left at default if not used.
- RoutableIPPoolCIDR requires five unused contiguous IP addresses in each range.

- The PFMPHostname value must be fully qualified (FQDN).
- PFMPHostIP value must be in the flex-node-mgmt-<vlanid> range.

6

Configuring PowerFlex Manager

After installing the PowerFlex management platform, perform the tasks in this section to complete the initial PowerFlex Manager configuration procedures. PowerFlex Manager is the web user interface for the PowerFlex management platform.

After you configure PowerFlex Manager, the remaining PowerFlex installation and configuration tasks are performed using PowerFlex Manager.

Log in to PowerFlex Manager

Perform the following steps to log in to PowerFlex Manager for the first time in order to continue configuration activities and deploy storage resources.

Prerequisites

- Ensure that you have access to a web browser that has network connectivity with PowerFlex Manager.
- Ensure that you know the address that was configured for accessing PowerFlex Manager. This address was configured for `PFMPHostname` in the JSON file during PowerFlex Manager installation.
- Prepare a new password for accessing PowerFlex Manager. The default password cannot be considered as a new password. For more information about default credentials, see the *Dell PowerFlex Family 5.x Security Configuration Guide*.

  Password rules are as follows:

  - Fewer than 32 characters
  - Only alphanumeric and punctuation characters

NOTE: If you are using **Windows Active Directory (AD)** as an **LDAP server**, you must NOT use hostnames (NetBIOS names) longer than **15** characters. Hostnames exceeding this limit may cause issues during deployment and integration.

Steps

1. Log in to PowerFlex Manager using the default credentials. The default credentials are provided in the *Dell PowerFlex Family 5.x Security Configuration Guide*.
2. Change the default password.

   Make a note of the password for future use. This password is for the SuperUser who is performing the initial configuration activities. Additional users and passwords can be configured later in the process.
3. Click **Submit**.
   You are now logged into the system. The initial configuration wizard is displayed. On the **Welcome** page, read the instructions, and click **Next** to proceed with initial configuration activities.

Perform the initial setup

The first time you log in to PowerFlex Manager, you are prompted with an **Initial Configuration** wizard, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Before you begin, have the following information available:

- Dell Technologies connectivity services configuration details
- Information about whether you intend to use a compliance file
- Information about the type of installation you want to perform, including details about your existing PowerFlex instance, if you intend to import from another PowerFlex instance

To configure the basic settings:

1. On the **Welcome** page, read the instructions and click **Next**.
2. On the **SupportAssist** page, optionally enableDell Technologies connectivity services and specify Dell Technologies connectivity services connection settings, and click **Next**.

NOTE: If you want to configure secure connect gateway and policy manager, complete *Enabling Dell Technologies connectivity services* before proceeding as enabling Dell Technologies connectivity services is an optional procedure and only applicable if you opt for it within your service agreement.

3. On the **Installation Type** page, specify whether you want to deploy a new instance of PowerFlex or import an existing instance, and click **Next**.
4. On the **Summary** page, verify all settings for Dell Technologies connectivity services, compliance, and installation type. Click **Finish** to complete the initial setup.

After completing the initial setup, you can begin to configure PowerFlex Manager and deploy resources from the **Getting Started** page.

Enabling Dell Technologies connectivity services for the PowerFlex management cluster (KVM)

There are two options to configure events and alerts:

- Connect directly
- Connect using secure connect gateway
- If you connect directly, only the call home option is available
- If you connect through secure connect gateway, all options through secure connect gateway are enabled

Deploy and configure secure connect gateway and policy manager using QCOW2 images

Use this procedure to deploy and configure secure connect gateway and policy manager using QCOW2 images.

Steps

1. Download the policy manager and secure connect gateway Hyper-V VHD files from Dell Technologies Support.
2. Log in to the KVM host as **delladmin**.
3. Download the VHD files to the KVM host (to `/tmp`).
4. Unzip the secure connect gateway VHD image, type: **unzip <filename>**.
5. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> scgvm.qcow2**.
6. Unzip the POLICYMGR VHD image, type: **unzip <filename>**.
7. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> policymgrvm.qcow2**.
8. Move the resulting QCOW2 images to `/mnt/general`, type: **mv *.qcow2 /mnt/general**.
9. Create VMs, type: **virt-install**:

```
sudo virt-install --name scgvm --ram 16384 --disk /mnt/general/scgvm.qcow2 --import
--vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --noautoconsole
--print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5901 > /tmp/scgvm.xml

sudo virt-install --name policymgr --ram 16384 --disk /mnt/general/policymgrvm.qcow2
--import --vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --
noautoconsole --print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5902 > /tmp/
scgvm.xml
```

10. Add crm configuration to instantiate both VMs as cluster objects, type:

```
sudo crm configure primitive scgvm VirtualDomain params config="/mnt/general/
scgvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta allow-
migrate=true target-role=Stopped op monitor interval=10s timeout=30s

sudo crm configure colocation co-general-scgvm info: scgvm cl-sharedfs-general
```

```
sudo crm configure order or-general-scgvm Mandatory: cl-sharedfs-general scgvm

sudo crm configure primitive policymgrvm VirtualDomain params config="/mnt/general/
policymgrvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta
allow-migrate=true target-role=Stopped op monitor interval=10s timeout=30s

sudo crm configure colocation co-general-policymgrvm info: policymgrvm cl-sharedfs-
general

sudo crm configure order or-general-policymgrvm Mandatory: cl-sharedfs-general
policymgrvm
```

11. Start the resources, type: **sudo crm resource start scgvm policymgrvm**.
12. Temporarily add the firewall rule to host to allow the VNC communication, type: **sudo firewall-cmd --add-port=5901/tcp --add-port=5902/tcp**.
13. Use the VNCviewer from the jump server (or laptop connected to service port) to connect to the console of scgvm, see Secure Connect Gateway 5.x — Virtual Edition Deployment Guide for the secure connect gateway and policy manager deployment procedures.
14. Clean up the firewall, type: **sudo firewall-cmd --remove-port=5901/tcp --remove-port=5902/tcp**
15. Clean up the downloaded files (`zip` and `VHD` files), type: **rm ~/POLICYMGR_* ; rm ~/SCG_***.

Configure secure connect gateway and policy manager

Use this procedure if you need to configure secure connect gateway and policy manager.

Steps

1. Connect to jump server and launch a console.
2. In a terminal on the jump server, type the following: **virt-viewer -a -c qemu+ssh://delladmin@<KVM Host>/system**.

   A window displays listing the VMs running on the host.
3. Select the secure connect gateway / policy manager VM to launch the console.
4. Agree terms, type **ALT** + **A** and type **ALT** + **N** to go next.
5. Edit the network settings, type **ALT** + **I**.
   a. Provide the IP address and subnet mask, type **ALT** + **N**.
   b. Provide the hostname, DNS and domain search, type **ALT** + **S**, and type **ALT** + **N** to go next.
   c. Provide routing, type **ALT** + **U**, and to add route, type **ALT** + **D**.
   d. Provide the gateway and select device as eth0, **ALT** + **O** and type **ALT** + **N** to go next.
   e. Select time zone and type **ALT** + **N** to go next.
   f. Provide the root password and type **ALT** + **N** to go next.
   g. Maintain SSL as default, and type **ALT** + **N** to go next.

      NOTE: This step is only for Policy Manager (PM).

   h. Maintain the user admin as default and type **ALT** + **N** to go next.
6. Wait for the configuration to complete. Once the configuration is complete access web UI for further configuration.
7. Repeat steps 3 to 6 to configure policy manager.

Register and log in to the secure connect gateway user interface

Use this procedure to log in to the secure connect gateway user interface.

Steps

To log in to the secure connect gateway, see *Registering and signing in to Secure Connect Gateway* in the Secure Connect Gateway 5.x — Virtual Edition Deployment Guide.

Ensure you complete the following procedures:

- *Sign in to Secure Connect Gateway*
- *Register Secure Connect Gateway*
- *Sign in using administrator account or network credentials*

Configure policy manager settings on secure connect gateway

Use this procedure to ensure connectivity between secure connect gateway and policy manager.

About this task

NOTE: Policy manager is not managed by PowerFlex Manager.

Steps

1. Sign in to secure connect gateway.
2. Go to **Settings** > **Environment configuration** > **Connectivity details** > **Policy Manager**.
3. Click **Enable remote Policy manager**.
4. Enter the hostname or IP address, port number, username, and password of the policy manager server.

   NOTE: If the port is SSL secured, the port number must be **8443**. If the port is not SSL secured, the port number must be **8888**.
5. Click **Enable SSL** if policy manager for secure connect gateway is installed on a server that is secured by SSL.
6. If the policy manager server connects to the Internet through a proxy server, perform the following steps in the **Customer proxy server** section:
   a. Click **Enable proxy server for Policy manager only**.
   b. Enter the hostname or IP address and port number.
   c. Click **Proxy requires authentication** and enter the username and password to access the proxy server.
7. Test the connection and click **Apply**.

Enabling Dell Technologies connectivity services for VMware ESXi

There are two options to configure events and alerts:

- Connect directly
- Connect using secure connect gateway
- If you connect directly, only the call home option is available
- If you connect through secure connect gateway, all options through secure connect gateway are enabled

Deploy and configure secure connect gateway

Secure connect gateway is an enterprise monitoring technology that monitors your devices and proactively detects hardware issues that may occur.

Prerequisites

Ensure your system complies to the minimum requirements, for more information, see the *Secure Connect Gateway 5.x - Virtual Edition Deployment Guide*.

Steps

To deploy and configure secure connect gateway, see Deploy Secure Connect Gateway using VMware vSphere Web Client in the *Secure Connect Gateway 5.x - Virtual Edition Deployment Guide*.

Register and log in to the secure connect gateway user interface

Use this procedure to log in to the secure connect gateway user interface.

Steps

To log in to the secure connect gateway, see *Registering and signing in to Secure Connect Gateway* in the Secure Connect Gateway 5.x — Virtual Edition Deployment Guide.

Ensure you complete the following procedures:

- *Sign in to Secure Connect Gateway*
- *Register Secure Connect Gateway*
- *Sign in using administrator account or network credentials*

Deploy policy manager for secure connect gateway

Use this procedure to deploy policy manager for secure connect gateway.

About this task

NOTE: Policy manager for secure connect gateway is optional.

PowerFlex Manager does not manage policies or policy manager for secure connect gateway.

This chapter does not cover procedures to create and manage policies. For information on creating and managing policies, see *Policy Manager for Secure Connect Gateway 5.00.00.xy User's Guide*.

Prerequisites

Before deploying policy manager for secure connect gateway, ensure the following:

- Upgrade secure connect gateway
- Download the latest version of policy manager for secure connect gateway from Dell Technologies support site.

Steps

1. Download and extract the OVF file to a location accessible by VMware vSphere Client.
2. On the right pane of VMware vSphere Client, click **Create/Register VM**.

   The **New virtual machine** window is displayed.
3. On the **Select creation type** page, select **Deploy a virtual machine from an OVF or an OVA file** and click **Next**.
4. On the **Select OVF and VMDK files** page, enter a name for the virtual machine, select the OVF and VMDK files, and click **Next**.

   The **Select storage** page is displayed. If there is more than one datastore on the host, the datastores are displayed on the **Select storage** page.
5. Select the location to store the virtual machine files and click **Next**.
6. Review the details and click **Next**.
7. On the **License agreements** page, read the license agreement, click **I agree**, and click **Next**.
8. On the **Select Storage** page, select the datastore for the VM, select a disk provisioning type, and click **Next**.
9. On the **Deployment options** page, perform the following steps:
   a. From the **Network mappings** list, select the network that the deployment template must use.
   b. Click **Next**.
10. On the **Additional settings** page, enter the following details and click **Next**:
    - Domain name server
    - Hostname
    - Default gateway
    - Network IPv4 and IPv6
    - Time zone
    - Secure socket layer

- Root password: Ensure that the root password consists of eight characters with at least one uppercase and one lowercase letter, one number, and one special character.
- Web administrator username: After deployment, the username is automatically updated to admin.

11. On the **Ready to complete** page, verify the details that are displayed and click **Finish**.
    A message is displayed after the deployment is complete and the virtual machine is powered on.
12. Log in to policy manager for secure connect gateway and perform the following steps:
    a. Go to **`https://<hostname(FQDN) or IP address>:8443`**
    b. Enter admin as the user ID.
    c. If you are logging in for the first time, enter the default password.
    d. After logging in for the first time, enter the new administrator account password when prompted and click **Save**.
    e. Enter the new password and click **Log In**.

Configure policy manager settings on secure connect gateway

Use this procedure to ensure connectivity between secure connect gateway and policy manager.

About this task

NOTE: Policy manager is not managed by PowerFlex Manager.

Steps

1. Sign in to secure connect gateway.
2. Go to **Settings** > **Environment configuration** > **Connectivity details** > **Policy Manager**.
3. Click **Enable remote Policy manager**.
4. Enter the hostname or IP address, port number, username, and password of the policy manager server.
   NOTE: If the port is SSL secured, the port number must be **`8443`**. If the port is not SSL secured, the port number must be **`8888`**.
5. Click **Enable SSL** if policy manager for secure connect gateway is installed on a server that is secured by SSL.
6. If the policy manager server connects to the Internet through a proxy server, perform the following steps in the **Customer proxy server** section:
   a. Click **Enable proxy server for Policy manager only**.
   b. Enter the hostname or IP address and port number.
   c. Click **Proxy requires authentication** and enter the username and password to access the proxy server.
7. Test the connection and click **Apply**.

Configuring Dell Technologies connectivity services on PowerFlex Manager

Depending on the customers requirement, use the following procedures to configure the connect directly or connect using the secure connect gateway. In order to allow remote support, use secure connect gateway. Remote support is not be enabled for direct connect.

Configure Dell Technologies connectivity services using the connect directly mode

Use this procedure to enable Dell Technologies connectivity services using the connect directly mode.

Steps

1. Log in as **`admin`** to PowerFlex Manager.
2. Click **Settings** > **Events and alerts**.

3. Click **Notification Policies**.
4. On the **Destinations** pane, click **Dell Technologies connectivity services**.
5. Accept the license and telemetry agreement on the connect support assist page and click **Next**.
6. Choose the connection type **Connect Directly**.

   NOTE: This helps us directly connect to Dell Technologies connectivity services direct. Call to home feature works on connect direct. The proxy setting is not supported.

7. Click **Connect to Dell AIOps**.

   It enables PowerFlex Manager to transport telemetry data, alerts and analytics to assist Dell Technologies in providing support.
8. On the **Authentication details** page, provide the following details.
9. Access key and PIN, see Generate Access Key and PIN for Secure Connect Gateway 5.x.

   You need the software ID for generating the access key and PIN.
10. Choose the **Device type** to be registered like rack, appliance or software.
11. In the **Enterprise License Management Systems** file, enter the software ID used in step 2 while generating the access key and pin.
12. The **Solution serial number** must be provided by customer.
13. In the **Site ID** field, provide the site ID location. If you do not have one, contact Dell Technologies Support to generate one.
14. Click **Next**, provide the contact details for customer, and click **Finish**.

    A window appears on the bottom of the screen configuring Dell Technologies connectivity services and another window appears once it is successfully configured.
15. To activate the policy, click **Configure Now** and enable the policy by making it active.

    Once the policy is active, the policy will move from grayed out mode to available and active mode.

Connect Dell Technologies connectivity services using the secure connect gateway

Use this procedure to enable Dell Technologies connectivity services using the secure connect gateway.

Prerequisites

Configure the secure connect gateway.

Steps

1. Log in as **admin** to PowerFlex Manager.
2. Click **Settings** > **Events and alerts**.
3. Click **Notification Policies**.
4. On the **Destinations** pane, click **Dell Technologies connectivity services**.
5. Accept the license and telemetry agreement on the **Connect to Dell Technologies connectivity service** page and click **Next**.
6. Choose the connection type **connect via gateway**.

   NOTE: Connect using the gateway helps register PowerFlex Manager on secure connect gateway and Dell Technologies connectivity services. From here we can enable the proxy setting.

7. Provide the **SCG IP address and Port** number.

   NOTE: Maintain the gateway port as 9443. Port 5700 port is only used for the secure connect gateway UI, API communication with secure connect gateway from the PowerFlex management platform uses port 9443.

8. Click **Connect to Dell AIOPs**.

   It enables PowerFlex Manager to transport telemetry data, alerts, and analytics to assist Dell Technologies in providing support.
9. Enable the **Remote Support** button and click **Next**.
10. On the **Authentication Details** page, provide the following details.

11. Access key and PIN, see Generate Access Key and PIN for Secure Connect Gateway 5.x.
12. Choose the **Device** type to register like rack, appliance, or software.
13. In the **Enterprise License Management Systems** file, enter the software ID used in step 4 while generating the access key and PIN.
14. The **Solution serial number** must be provided by the customer.
15. In the **Site ID** field, provide the site ID location. If you do not have one, contact Dell Technologies Support to generate one.
16. Click **Next**, provide the contact details for the customer, and click **Finish**.

    A popup appears on the bottom of the screen configuring Dell Technologies connectivity services and another pop up appears once it is successfully configured.
17. To activate the policy now, click **Configure Now** and enable the policy by making it active.

    Once the policy is active, it will remove from grayed out mode to available and active mode.
18. To verify the registration, check the PowerFlex management platform log to monitor registration status, type: **`= kubectl get supportassist -n powerflex -o yaml`**.

    Example of a successful registration:

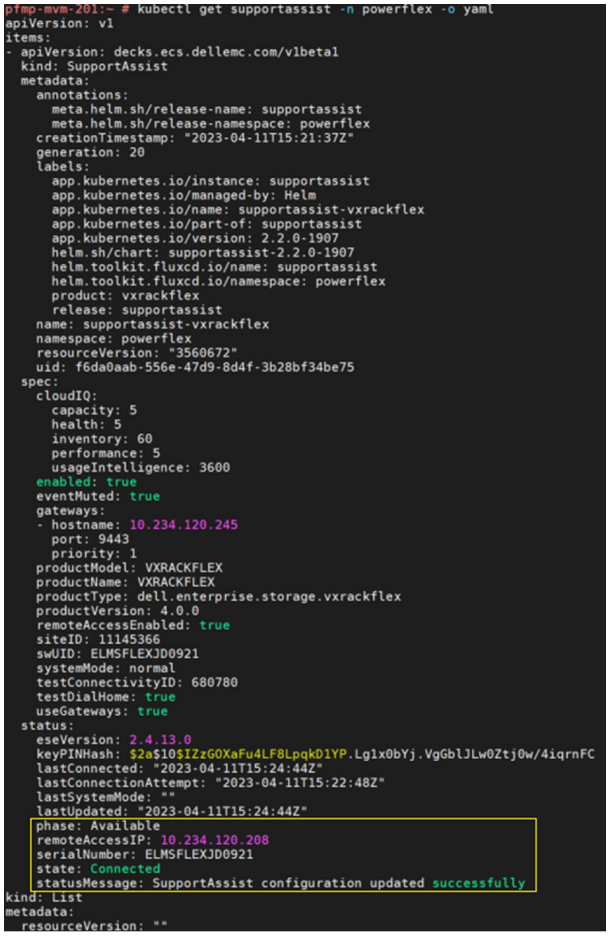


Figure 3. Successful registration of the PowerFlex management platform

Verify secure connect gateway status

Perform this procedure to check the status of the secure connect gateway.

Steps

1. Using a web browser, access secure connect gateway by typing the following URL: **`https://<scg-gateway_ip>:5700`**
2. On the menu bar, select **Device Management** > **Managed Devices** > **View**. Initially the device displays as **Offline** or **Missing**.

3. After 10 to 20 minutes, the device displays as **Online** and **Managed**.

7

Deploying PowerFlex storage

The next stage of installation involves defining infrastructure and resources, and then deploying PowerFlex storage.

Deploy the PowerFlex data storage as a resource group by using one of the following methods:

- Deploying a PowerFlex cluster using a CSV topology file from PowerFlex Manager.
- Deploy a PowerFlex cluster using REST API commands.

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Deployment design for integrated PowerFlex management (non-KVM)

The following deployment design principles are recommended:

- If the hardware requirements are met, any node running a storage node can also optionally run an SDT.
- Ensure that the LIA is installed on every node, as it is required for maintenance purposes.

PowerFlex component layout

The following table details best practices in relation to component layout for a co-resident deployment, based on the number of storage nodes you are deploying:

Table 16. PowerFlex component layout (2 + 2)

Number of nodes	2+2 (Minimum five nodes) Components installed on the node							
Node 1	**Node 2**	**Node 3**	**Node 4**	**Node 5**	**Node 6**	**Node 7**	**Node 8**	
Five nodes	Primary MDM Storage node PowerFlex management platform	Secondary MDM Storage node PowerFlex management platform	Secondary MDM Storage node PowerFlex management platform	Tiebreaker Storage node PowerFlex management platform	Tiebreaker Storage node PowerFlex management platform	N/A	N/A	N/A
Six nodes	Primary MDM Storage node	Secondary MDM Storage node PowerFlex management platform	Secondary MDM Storage node PowerFlex management platform	Tiebreaker Storage node PowerFlex management platform	Tiebreaker Storage node PowerFlex management platform	Storage node PowerFlex management platform	N/A	N/A
Seven nodes	Primary MDM	Secondary MDM	Secondary MDM	Tiebreaker	Tiebreaker	Storage node	Storage node	N/A

Table 16. PowerFlex component layout (2 + 2) (continued)

Number of nodes	2+2 (Minimum five nodes) Components installed on the node							
	Storage node	Storage node	Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform	PowerFlex management platform	PowerFlex management platform	
Eight nodes	Primary MDM Storage node	Secondary MDM Storage node	Secondary MDM Storage node	Tiebreaker Storage node PowerFlex management platform	Tiebreaker Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform

Table 17. PowerFlex component layout (8 + 2)

Number of nodes	8 + 2 (Minimum eleven nodes) Components installed on the node										
Eleven nodes or higher	**Node 1**	**Node 2**	**Node 3**	**Node 4**	**Node 5**	**Node 6**	**Node 7**	**Node 8**	**Node 9**	**Node 10**	**Node 11**
	Primary MDM Storage node	Secondary MDM Storage node	Secondary MDM Storage node	Tiebreaker Storage node	Tiebreaker Storage node	Storage node	Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform	Storage node PowerFlex management platform

NOTE: For systems with additional storage nodes, separate the PowerFlex management platform services from the MDM roles, if possible.

The following table describes the PowerFlex MDM certificates that must be installed on the participating nodes for SUSE Linux Enterprise Server before starting a integrated PowerFlex management (non-KVM) deployment.

Table 18. PowerFlex MDM certificates for SUSE Linux Enterprise Server

For PowerFlex MDM certificates	Location	Repository
pycparser-2.21-py2.py3 or higher	Third-party	Python Package Index
cffi-1.15.1 or higher	Third-party	Python Package Index
cryptography-40.0.2 or higher	Third-party	Python Package Index

Download the appropriate versions of pycparser, cffi and cryptography for your operating computer release and Python version. SLES 15 SP6 and above comes with these modules installed.

The following table describes the PowerFlex MDM certificates that must be installed on the participating nodes for Red Hat Enterprise Linux before starting a integrated PowerFlex management (non-KVM) deployment.

Table 19. PowerFlex MDM certificates for Red Hat Enterprise Linux

For PowerFlex MDM certificates	Location	Repository
pycparser-2.21-py2.py3 or higher	Third-party	Python Package Index
cffi-1.15.1 or higher	Third-party	Python Package Index
cryptography-40.0.2 or higher	Third-party	Python Package Index

Download the appropriate version of pip, pycparser, cffi and cryptography for your operating computer release and Python version. Python v3 and Python3-cryptography includes the above mentioned modules and can be used to run the MDM certificate generation script. RedHat 9 and above will already include them.

Operating system requirements

The following table lists the required packages for each operating system for PowerFlex 5.0.x. A valid time source should be configured on all nodes before deploying PowerFlex.

For the most updated list of operating systems supported by PowerFlex, see *Dell PowerFlex 5.0.x Support Matrix* on the Dell Technologies support site for the supported versions.

Linux

Table 20. Linux requirements

Component or feature	Required packages
All components, all Linux types	• numactl • libaio • wget • apr • libapr • python2-rpm or python3-rpm • yum-utils
MDM	• bash-completion (for SCLI completion) • Latest version of Python 2.x or Python 3.x (for newer operating system such as Red Hat Enterprise Linux 8.x) • binutils • java-11-openjdk-headless • java-17-openjdk-headless
PowerFlex data server	• smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils
Diagnostic tools	• sar • sysstat
Storage controllers	• DELL: perccli • LSI raid controllers: storcli • HP raid controllers: hpssacli
Storage Nodes that support SDPMs or NVDIMMs	• ndctl • jq • daxio • libpmem
Secure authentication mode	Ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system: • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm`
LDAP	Ensure that OpenLDAP 2.4 is installed on all servers
LIA	pciutils

VMware ESXi

PowerFlex only supports the SDC component VMware ESXi 8.x for compute only nodes (SDC).

Microsoft Windows

PowerFlex only supports the SDC component on Microsoft Windows.

Table 21. Microsoft Windows requirements

Operating system	Requirement
Windows	• Only SDC is a supported component on Windows. • To install SDC, ensure that Security Update for x64-based Systems is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported) • Visual C++ redistributable 2010 package, 64-bit The Microsoft Windows *Storage Spaces* feature (available from Windows 8.1 and later) is not supported in Microsoft failover cluster running on PowerFlex volumes.

Prepare Linux-based server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 storage data client, use this procedure to prepare the Linux-based operating systems of servers that host the PowerFlex metadata manager, and storage node components.

Prerequisites

Superuser permissions are required to configure the servers.

About this task

Configure the servers hosting MDM, and storage node to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting the MDM, and the storage node.
- For all servers hosting the MDM, and storage node components, certain threshold parameters must be configured. This change enables a larger number of records in the Address Resolution Protocol (ARP) cache table of the operating system.

Steps

1. On servers hosting the PowerFlex metadata manager, and Storage Node, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that the operating system can open.
   b. If the number is 32768 or higher, go to step 2. If the number is less than 32768, go to the next substep.
   c. Open the file `/etc/sysctl.conf` using the vi editor.
   d. If the following line exists, edit it with the vi editor. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 32768
      ```

2. On servers hosting the MDM, and storage node, if the following lines exist in the file `/etc/sysctl.conf`, edit with the vi editor. If they do not exist, add the full lines with the following settings:

```
net.ipv4.neigh.default.gc_thresh1 = 8192
net.ipv4.neigh.default.gc_thresh2 = 16384
net.ipv4.neigh.default.gc_thresh3 = 32768
```

3. Save the file.
4. Apply the changes, using the command:

```
sysctl -p
```

5. Repeat the steps on the next server, until all servers are configured.

Configuring SDPM or NVDIMM devices for PowerFlex storage-only nodes

Use the following procedures in this section to configure SDPM or NVDIMM devices for PowerFlex storage-only nodes.

NOTE: It is recommended not to mix SDPM and NVDIMM in the same protection domain.

- Verify that the embedded operating system is installed and that the network is configured.
- Verify that the PowerFlex storage-only node meets the requirements using configuring storage-only nodes with SDPMs or NVDIMMs.

NOTE: If you are configuring PowerFlex storage-only nodes without SDPM or NVDIMM, you can skip this section.

Modifying SDPM settings and sanitizing SDPM with iDRAC

Use this procedure to configure SDPM using iDRAC.

About this task

Based on the capacity of the node, this procedure can be used to modify the SDPM settings. It is applicable only to PowerFlex R660 and R760 nodes.

NOTE: This applies only to nodes with SDPM enabled, where the customer intends to increase the default value.

For PowerFlex R660 and R760 nodes, the SDPM page can be accessed directly through the iDRAC interface.

Steps

1. Log in to iDRAC using your administrator credentials.
2. Go to the **Configuration** tab.
3. Select **BIOS Settings** from the left-hand menu.
4. Expand the **Memory Settings** section and click on **Persistent Memory**.
5. Under the **Software Defined Persistent Memory** section, locate the **SDPM Default Memory Size** field.
6. Update the value based on the recommended size provided in the table *Memory sizes*.
7. To sanitize the SDPM:

   NOTE: Enabling the sanitize option will permanently erase all data stored on the SDPM.

   a. Log in to iDRAC.
   b. Go to the **Configuration tab** and select **BIOS Settings**.
   c. Expand **Memory Settings** and click on **Persistent Memory**.
   d. Locate and enable the **Sanitize ALL NVDIMMs** option.
   e. Click **Apply** and **Reboot** the system to initiate the sanitization process.

Enable SDPM in the BIOS

About this task

NOTE: SDPM is available only on PowerEdge R660 and R760 servers only. This section describes how to prepare SDPM for use as an acceleration device. This option is only available on the servers where SDPM is enabled from the factory.

Steps

1. Open a web browser to the iDRAC IP address and log in.
2. From the **Dashboard**, click **Virtual console** to open a session.
3. Start the KVM session.
4. Click **Boot** and select **BIOS setup**, then click **Yes**.
5. Click **Power** and select **Power On system**.
6. Select **System BIOS** > **Memory Settings** > **Persistent Memory** > **Software Define Persistent Memory**.
7. Set the memory size according to the chart below:

Table 22. Memory sizes

Node storage capacity (GB)	Required SDPM capacity (GB)
less than 512	64
Equal to 512	128

8. Click **Back**.
9. Select **Enabled** for **Sanitize all SDPMs**.
10. Click **OK** on the **SDPM data will be erased** warning box.
11. Click **Back** > **Back** > **Finish** then **Yes** to confirm.
12. Click **OK** on the **Settings were saved successfully** box.
13. Click **Finish** to exit the system setup menu and **Yes** to confirm.

Verify the PowerFlex version

Use this procedure to verify the installed PowerFlex version.

Steps

1. Log in to the jump server.
2. SSH to primary MDM.
3. Log in with administrator credentials.
   ```
   scli --login --username admin --password 'admin_password' --management_system_ip <powerflex_manager_ip>
   ```
4. Type `scli --version` to verify the PowerFlex version.

Verify the operating system version

Use this procedure to verify that embedded operating system is installed.

Steps

1. Log in the jump server.
2. SSH to the PowerFlex storage-only node.
3. Enter the following command to verify the operating system version:
   ```
   cat /etc/*-release
   ```

Verify necessary RPMs

Use this procedure to verify necessary RPMs are installed.

About this task

The following RPMs must be installed:

- ndctl
- ndctl-libs
- libpmem
- libpmemblk

The sample output in this procedure is for reference purpose only. The versions may vary depending on the used.

Steps

1. Log in to the jump server.
2. SSH to the PowerFlex storage-only node.
3. Type **`zypper search-packages ndctl ndctl-libs libpmem libpmemblk`**
4. If the RPMs are not installed, type **`zypper install <rpm>`** to install the RPMs.

Create namespaces for SDPM or NVDIMM devices

Use this procedure to create namespaces, and configure each namespace for an SDPM or NVDIMM device.

Steps

1. SSH to the PowerFlex storage-only node.
2. For each SDPM, type (starting with **namespace0.0**):

   `ndctl create-namespace -m fsdax`

   For example:

   ```
   {"dev":"namespace0.0","mode":"fsdax","map":"dev","size":"15.75 GiB
    (16.91 GB)","uuid":"348d510e-dc70-4855-a6ca-6379046896d5","raw_uuid":
   "4ca5cda2-ebd4-4894-aa4e-0cfc823745e2","daxregion":{"id":0,"size":"15.75 GiB (16.91
   GB)","align":4096,"devices":[{"chardev":"dax0.0","size":"15.75 GiB (16.91 GB)"}]},
   "numa_node":0}
   ```

3. Repeat for each SDPM or NVDIMM namespace.

Create the persistent memory

Use this procedure to create the persistent memory device.

Steps

1. Log in to the PowerFlex storage-only node (primary MDM).
2. Using SCLI, connect to the PowerFlex cluster:

   `scli --login --username admin --management_system_ip <PFXM-ui-IP> --password <PFXM-ui-password>`
3. Type **`scli --add_device_group --protection_domain_name PD_EC --media_type PMEM --device_group_name pmem_ec`** to create the device group with the persistent memory device type.

Add the FSDAX devices to the storage node

Add the newly created FSDAX devices to the storage node.

Prerequisites

Ensure that you have the IP address and admin login credentials for accessing PowerFlex Manager. If necessary, the customer can provide you with the necessary information.

Steps

1. Log in to the PowerFlex Manager with the credentials.
2. On the menu bar, go to **Lifecycle** > **Storge Nodes**.
3. Select the storage node that is in maintenance mode, click **View Details**
   a. Click **Add Device**.
   b. In the **Add Device to Storage Node** dialog box, enter the relevant information as below.
      - **Path** – **`/dev/pmemX`**. Use the path that is created during FSDAX device.
      - **Name** – Use the path name as the name.
      - **Device Group** – Select the pmem device name from the drop-down list. The device group name is recorded in the table.

        NOTE: You notice the media type as PMEM once you select the device group.

   c. To add more than one device, click **Add row** and repeat the step b.
   d. Expand **Advanced Options**, move the enable icon for **Force Device Takeover**. Click **Add devices**.

Deploying PowerFlex using a non-root user

Install or expand PowerFlex in Linux environments using a nonroot sudo user in noninteractive mode. This option is used only when a CSV file is used to install PowerFlex.

Sudo is a program that allows you to run or install a program as the root user.

To deploy PowerFlex with a nonroot user, the nonroot user must meet the following conditions:

- The username in the CSV file must already exist.
- A nonroot user must be a sudo user.
- Nonroot user must be in noninteractive mode.

In the CSV file used for installation, you must indicate that you are intending to use a sudo nonroot username by appending the string **`(sudo)`** to the username in the **Username** field. For example, if you are using a nonroot user with the username *non_root*, enter the string **`non_root(sudo)`** in the username field of the CSV file.

Configure a non-root non-interactive sudo user

In Linux, you can install or extend PowerFlex with a nonroot user when installing PowerFlex using a CSV file.

About this task

The following procedure describes one method for configuring a nonroot noninteractive sudo user. Perform the commands from the operating system console of the node where PowerFlex will be installed.

Steps

1. Open the sudoers `/etc/sudoers` file for editing.

   Delete the following:

```
"Defaults targetpw"
ALL ALL=(ALL) ALL
```

Exit the vi editor, type the following command: **:wq!**.

2. Open the sudoers common criteria file for editing

```
vi /etc/sudoers.d/common-criteria
```

Modify the content to the following:

```
%trusted ALL=(ALL) NOPASSWD: ALL
```

3. Exit the vi editor by typing the following command to exit, enter the following: **:wq!**

Deploying a PowerFlex cluster using a CSV topology file

This section describes how to prepare a CSV topology file, upload the CSV topology file using REST API commands or PowerFlex Manager, and then deploy block storage using REST API commands or PowerFlex Manager.

If you are using the legacy PowerFlex to provide an installation topology file using PowerFlex Manager, click **Deploy With Installation File**.

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Consider the following:

- While deploying PowerFlex using PowerFlex installer, if you see an error indicating that the channel cannot be opened, then verify if PowerFlex is already deployed.
- While deploying additional servers with SDTs on an existing system with components such as SDTs, you may see errors listed in the **Events** tab. These errors are for the already existing components, such as, SDTs in the cluster.
- Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

For more information, see PowerFlex: On-Demand Compilation Of The PowerFlex SDC Driver.

Prepare the CSV topology file

Create a CSV topology file when a CSV file and REST API commands are used to deploy block storage. Follow the instructions that match your environment and the PowerFlex support matrix for your version. CSV file templates are provided as part of the software download.

Prerequisites

NOTE: PowerFlex 5.0.x only supports one storage pool, one device group and one persistent memory device group per protection domain.

Ensure that the settings you enter comply with PowerFlex product limits. See the Dell PowerFlex Technical Overview for product limits.

Ensure that you have a copy of the following CSV templates. Select a template that is most suitable for your requirements. In this explanation, the CSV file is referred to and illustrated as a spreadsheet.

Table 23. CSV templates

CSV template	Description
`PowerFlex_config_example_all fields.csv` (All fields)	This spreadsheet template contains all available fields, both required and optional. In particular, it contains an optional section for storage pool configuration, which is relevant for multiple nodes.
`PowerFlex_basic_example_minimum fields.csv` (Minimum fields)	This spreadsheet template contains only the required fields. The optional fields (those that are in the all fields spreadsheet, but not in the minimum fields spreadsheet) will be assigned default values.

NOTE: In two-layer topologies where frontend servers are Linux or Windows-based, SDCs must be represented by separate, additional rows. The values for SDC-specific rows are described in *CSV topology for two-layer deployment*. Alternatively, omit the SDCs from the CSV file, and install them manually after deploying the backend.

NOTE: Save a copy of the CSV file for future use. It is recommended to save the file in a secure encrypted folder on your operating system, after deployment. When you add nodes to the system in the future, you need the CSV file that you used for initial deployment of the system.

You can edit a CSV file with Microsoft Excel or another file-editing software.

Steps

1. In the CSV file, enter your site-specific information in the appropriate places. You can overwrite the default information provided in the file.

   You must only use one spreadsheet for the installation, as follows:

   - To manually enter all configuration details, use the all fields spreadsheet.
   - To use default values for the non-mandatory fields, use the minimum fields spreadsheet.
   - To configure non-default values for columns that are not in the minimum fields spreadsheet, either use the all fields spreadsheet, or copy the column heading from there into the minimum fields spreadsheet, and enter your custom values into the minimum fields spreadsheet.

   NOTE: The examples provided here do not show all possible parameters that are configurable in a PowerFlex system. The following figure shows the CSV files that are used:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Username	Password	Operating System	Is MDM/TB	MDM Mgmt IP	MDM Ips	MDM Name	perfProfileForMDM	Virtual IPs	Virtual IP NICs	is EC Service	EC Service Name
2	root	password	linux	Primary	172.24.10.177	172.24.10.177	MDM1	High	172.27.40.180	ens161	Yes	Node177
3	root	password	linux	Secondary	172.24.10.178	172.24.10.178	MDM2	High	172.27.40.180	ens161	Yes	Node178
4	root	password	linux	Secondary	172.24.10.179	172.24.10.179	MDM3	High	172.27.40.180	ens161	Yes	Node179
5	root	password	linux	TB	172.24.10.180	172.24.10.180	MDM4				Yes	Node180
6	root	password	linux	TB	172.24.10.181	172.24.10.181	MDM5				Yes	Node181
7												
8												
9												
10	Storage Pool Configuration											
11	ProtectionDomain	Device Grou	StoragePool	Data Layout	EC Storage pool size in GB	PMEM in use by SP	Protection Scheme					
12	PD_EC1	DG1	SP_EC1	EC	use_all	Yes	2+2					
13												
14	Device Group Configuration											
15	Protection Domain	Device Grou	Media Type	EC spare number of nodes								
16	PD_EC1	DG1	SSD	1								
17	PD_EC1	DG_pmem	PMEM									
18												

Figure 4. Sample CSV file

The following figure shows a sample of all fields in the CSV file, where SDCs are defined below the rows for MDMs and storage nodes. In a two-layer configuration. The Linux servers form the MDM cluster, and each of those servers will also be storage nodes. SDCs can be added here, in separate rows, or added to the system later.

Username	Password	Operating System	Is MDM/TB	MDM Mgmt IP	MDM Ips	MDM Name	perfProfileForMDM	Virtual IPs	Virtual IP NICs	is EC Service	EC Service Nam	EC IP Role All IP	Protection Domain	Device Group	Storage Device	Storage Device	Is SDC	perfProfileForSDC	PMEM Device L	Is SDT	SDT Name
root	password	linux	Primary	172.24.10.177	172.24.10.177	MDM1	High	172.27.40.180	ens161	Yes	Node177	172.24.10.177	PD_EC1	DG1	/dev/nvme0n1,	EC177-disk1,EC	No	Normal	/dev/dax0.0,/d	No	SDT177
root	password	linux	Secondary	172.24.10.178	172.24.10.178	MDM2	High	172.27.40.180	ens161	Yes	Node178	172.24.10.178	PD_EC1	DG1	/dev/nvme0n1,	EC178-disk1,EC	No	Normal	/dev/dax0.0,/d	No	SDT178
root	password	linux	Secondary	172.24.10.179	172.24.10.179	MDM3	High	172.27.40.180	ens161	Yes	Node179	172.24.10.179	PD_EC1	DG1	/dev/nvme0n1,	EC179-disk1,EC	No	High	/dev/dax0.0,/d	No	SDT179
root	password	linux	TB	172.24.10.180	172.24.10.180	MDM4				Yes	Node180	172.24.10.180	PD_EC1	DG1	/dev/nvme0n1,	EC180-disk1,EC	No	High	/dev/dax0.0,/d	No	SDT180
root	password	linux	TB	172.24.10.181	172.24.10.181	MDM5				Yes	Node181	172.24.10.181	PD_EC1	DG1	/dev/nvme0n1,	EC181-disk1,EC	No	High	/dev/dax0.0,/d	No	SDT181
root	password	linux		172.24.10.182	172.24.10.182					Yes	Node182	172.24.10.182	PD_EC1	DG1			yes				
root	password	linux		172.24.10.183	172.24.10.183					Yes	Node183	172.24.10.183	PD_EC1	DG1			yes				
root	password	linux		172.24.10.184	172.24.10.184					Yes	Node184	172.24.10.184	PD_EC1	DG1			yes				

Figure 5. SDCs defined in a sample CSV file

2. For Linux nodes, the **Domain** column is not relevant. Leave the column blank or remove the column.
3. There is only one storage pool per protection domain, and unless configured otherwise, it uses the entire space of the device group.
4. The default spare settings allocate one node as the spare.

   The following table describes the fields in the spreadsheets. The required fields appear in both all fields and minimum fields spreadsheets. Field names are case-sensitive; the order of the columns is not significant.

Table 24. Fields in the CSV file

Field	Description
IPs	IP address to be used for multiple purposes. Use this field to designate one IP address that will be assigned to all

Table 24. Fields in the CSV file (continued)

Field	Description
	the following: MDM IP, MDM Mgmt IP. This option is provided for use cases where separate networks for data and management are not required.
Username	The name of the user. ● Linux nodes: Enter either `root` or a non-root sudo user, indicated by appending **`(sudo)`** to the end of the username (default: `root`). For more information about deploying with a non-root user, see *Deploying PowerFlex using a non-root user*. ● Windows nodes: Enter a user with administrator rights (default: `administrator`).
Domain	If using a domain user, the name of the domain (not relevant for Linux).
Password	Password used to log in to the node. This should be the password of the user who is entered in the **Username** column.
Operating System	The operating system of the server: Linux or Windows.
Is MDM/TB	The MDM role to deploy on this node: `Primary`, `Secondary`, `TB`, `Standby-Secondary`, `Standby-TB`, or blank (if not an MDM). For more information, see the *Dell PowerFlex 5.0.x Technical Overview* for information about the MDM cluster.
MDM Mgmt IP	The IP address for the management-only network. The management IP address is not required for: tiebreaker, standby tiebreaker, and any host that is not an MDM. In such cases, leave this field blank.
MDM IPs	MDM IP addresses used to communicate with other PowerFlex components in the storage network. This is required for all MDMs, tiebreakers, and standbys. Leave this field blank for hosts that are not part of the MDM cluster. Maximum of eight addresses, comma-separated, no spaces.
MDM Name	The name to be assigned to an MDM node, up to 31 characters in length. MDM names must be unique and may not contain spaces. Ensure that each server on which MDM is installed has a unique hostname.
perfProfileForMDM	Optional performance profile to set for MDMs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature.
Virtual IPs	A virtual IP address (VIP) for each possible manager MDM. This virtual IP address can be used for communication between the MDM cluster and SDCs on the PowerFlex data networks (not applicable for the management network). Only one virtual IP address can be mapped to each NIC, with a maximum of four virtual IP addresses per system.
Virtual IP NICs	The NIC to which the virtual IP addresses are mapped. Maximum length of 11 characters.
Is EC Service	Deploy the EC component on this node: `Yes` or `No`.

Table 24. Fields in the CSV file (continued)

Field	Description
EC Service Name	The name for the storage node, up to 31 characters in length. The name must not contain spaces. Ensure that each server on which EC is installed has a unique hostname.
EC IP Role Backend IPs	EC IP addresses to be used for communication among all nodes. Comma-separated, no spaces. Storage only IPs - used for connection between PDS and DGWT, PDS and MDM, DGWT and MDM.
EC IP Role Frontend IPs	EC IP addresses to be used for communication among all nodes. Comma-separated, no spaces. Application only IPs - used for connection between SDC and PDS.
EC IP Role All IPs	EC IP addresses to be used for communication among all nodes. Comma-separated, no spaces. Storage and application IPs - used for all connection.
Device group	Set a device group to this storage node. Device group is relevant only for storage nodes.

Storage Device List	Storage devices to be added to a storage node. For more than one device, use a comma-separated list, with no spaces. Ensure that the device names and paths do not contain any special characters, such as: `?`, `` ` ``, `!`, `@`, `#`
	Device name format on Linux:
	`/dev/sdb,/dev/sdc`
	When specifying the storage node device path on a Linux node, use the path according to how it is listed in `cat/proc/partitions` (and not according to the output of `fdisk -l`). Use persistent device names.
	For example:
	`fdisk` output: `/dev/mapper/samah-lv1, /dev/sdb`
	`cat /proc/partitions` output: `dm-3, sdb`
	Use these values in the CSV file: `/dev/dm-3`, `/dev/sdb`
	Device data is erased when devices are added to an storage node. When adding a device to an storage node, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is displayed for each device.
	PowerFlex might not perform optimally if there are large differences between the sizes of the devices in the storage pool—for example, if one device is as big as the rest of the devices. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI command `modify_storage_node_device_capacity`.
	For optimal performance, try to balance the number of devices of a storage pool, and the capacity of those devices among the relevant storage nodes.

Is SDC	`Yes` or `No`.
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configure performance profile during deployment*.

Table 24. Fields in the CSV file (continued)

Field	Description
SDC Name	Sets a name for the SDC, with a maximum length of 31 characters. Ensure that each server on which SDC is installed has a unique hostname.
PMEM Device List	PMEM storage devices to be added to an storage node.
Is SDT	Deploy the SDT component on this node: `Yes` or `No` SDT is an NVMe target, which is a frontend component that translates NVMe over Fabric protocol into internal protocols.
SDT Name	The name for the SDT node can be up to 31 characters in length. The name must not contain spaces.
SDT Host IPs	It is the network interface that allows the host to interact with the PowerFlex cluster's MDM.
SDT Storage IPs	This IP is dedicated to storage traffic for reading and writing data to and from the storage devices.
SDT All IPs	SDT IP addresses used for both hosts communication and MDM communication (including both roles). Maximum of eight IP addresses, comma-separated, no spaces.
Storage Pool configuration	There is only one storage pool per protection domain, and unless configured otherwise, it uses the entire space of the device group - spare.
Protection Domain	The Protection Domain to which the storage pool belongs.
Storage Pool	The storage pool name.
Data Layout	Data layout type: EC - for EC storage pool
Compression Method	The compression method used in the fine granularity storage pool. Compression methods might affect performance. ● `None` - no compression is used ● `Normal` - compression is enabled
Device group	Set device group to storage pool.
EC storage pool size in GB	Set size to more than 8 GB to storage pool or `use_all`
PMEM in use by storage pool	PMEM is in use:`Yes` or `No`
Protection Scheme	The protection scheme for storage pool: 2 + 2 8 + 2
Device Group Configuration	
ProtectionDomain	The protection domain to which the device group belongs.
Device group	The device group name.

Media Type	The expected device media type in the device group:
	`SSD`
	`PMEM`

EC spare number of devices	Number of spare devices: 0 **NOTE:** Spare number of devices is not supported with PowerFlex 5.0.x.

Table 24. Fields in the CSV file

Field	Description
EC spare number of nodes	Minimum spare node requirement is 1. The range is between 1 and 2.

5. Save the CSV file and retain it for future use.

Prepare the CSV topology file for two-layer deployment

When deploying SDC nodes on a Linux server in a two-layer environment, add rows to the CSV file specifically for SDCs, that only contain the values listed in this table.

Table 25. CSV topology file for a two-layer deployment

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user Linux nodes: Enter either `root` or a `non-root` sudo user, indicated by appending "(sudo)" to the end of the username (default: `root`). For more information about deploying with a `non-root` user, see related information.	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column.	Yes
Operating System	The server's operating system: `linux` or	Yes
-		
Is SDC	`Yes`	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see the *Configuring performance profile during deployment topic* in related information.	-
SDC Name	Name for the SDC.	-

CSV file examples for typical configurations

Refer to the examples provided in this section to see how the CSV file is used to configure various system topologies and features.

NOTE: These examples represent a subset of all possible topologies and features that can be configured.

Four data networks: two front-ends and two back-ends

This example shows a CSV topology file for four data networks, where two are front-end networks, and two are back-end networks.

Four data networks are configured as follows:

- Two front-end networks for SDCs
- Two back-end networks for MDMs and Storage Node

To download the example CSV file, see the *Dell PowerFlex 5.0.x and PowerFlex Manager 5.0.x Example CSV files* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex Family 5.x Security Configuration Guide*.

Table 26. CSV file parameters

Parameter	Example
Username	non-root
Password	password
Is MDM/TB	Primary Other options: Secondary, tiebreaker (TB)
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
EC Service Name	storageNode1
Device Group	deviceGroup1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153,ens154
Is EC service	Yes
EC IP Role Backend IPs	192.168.151.11,192.168.152.11
EC IP Role Frontend IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
Is SDC	No
perfProfileForSDC	high
Storage Pool Configuration (Section is used to configure data layout)	
ProtectionDomain	PD01
StoragePool	SP01
Data Layout	EC
Compression Method	Normal
Device Group	deviceGroup1
EC Storage pool size in GB	Add a desired value of above 8GB. In the case of using a full pool use **`use_all`**
PMEM in use by SP	No
Protection Scheme	2+2
Device Group Configuration	
Protection Domain	PD01
Device Group	deviceGroup1
Media Type	SSD **NOTE:** If SDPM/NVDIMM is used, the media type is PMEM
EC spare number of devices	0
EC spare number of nodes	1

Four data networks: Front-end and back-end on all four networks

This example shows a CSV topology file for four data networks, where all four are both front-end and back-end networks.

Four data networks are configured as follows:

- Four networks for both front-end and back-end: SDCs, MDMs and storage nodes

To download the example CSV file, see the *Dell PowerFlex 5.0.x and PowerFlex Manager5.0.x Example CSV files* available on the Dell Technologies support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary. For information about default credentials, see the *Dell PowerFlex Family 5.x Security Configuration Guide*.

Table 27. CSV file parameters

Parameter	Example
Username	non-root
Password	password
Is MDM/TB	Primary Other options: Secondary, tiebreaker (TB)
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
EC Service Name	storageNode1
Device Group	deviceGroup1
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10
Virtual IP NICs	ens151,ens152,ens153,ens154
Is EC service	Yes
EC IP Roles	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
Is SDC	No
perfProfileForSDC	high
PMEM Device List	/dev/dax0.0
PMEM Pool List	accp01
Storage Pool Configuration (this section is used to configure data layout)	
ProtectionDomain	PD01
StoragePool	SP01
Data Layout	EC
Compression Method	Normal
Device Group:	deviceGroup1
EC Storage pool size in GB	8192
PMEM in use by SP	No

Table 27. CSV file parameters (continued)

Parameter	Example
Protection Scheme	2+2
Device Group Configuration	
Protection Domain	PD01
Device Group	deviceGroup1
Media Type	SSD **NOTE:** If SDPM is used, the media type is PMEM
EC spare number of devices	0
EC spare number of nodes	1

Configuring performance profile during deployment

During deployment, system performance is enhanced by default implementation of the high performance profile for the MDM, SDC and Storage Node components. Most servers support the requirements for this configuration and can benefit considerably from its implementation.

High performance profile should be enabled on servers equipped with processors based on Haswell-equivalent or newer architectures and 48 GB of RAM. Other configurations may benefit from this setting as well.

The performance profile is set to High by default, but if you want to change the profile to Compact in the CSV file, use the `perfProfileFor<COMPONENT>` columns.

Deploy a PowerFlex cluster using a CSV topology file from PowerFlex Manager

Use the following procedure to deploy a PowerFlex cluster using a CSV topology file.

Steps

1. Log in to PowerFlex Manager.
2. On the **Getting Started** page, click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.
3. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts (RPMs) according to the operating system of your PowerFlex operating system machines.
4. Click **Upload Files** and click **Next** when the RPMs have finished uploading.
5. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   To view sample files, click the links to the following two example files: **PowerFlex_config_example_all fields.csv (all fields)** and **PowerFlex_basic_example_minimum fields.csv (minimum fields).**
6. Click **Upload Installation CSV**.
7. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
8. Click **Next**.
9. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex 5.0.x. You can skip options or force an overwrite.
10. Select **Force overwrite of existing data on devices used for PowerFlex storage**.

    PowerFlex can be installed on new or clean disks (storage node-DGWT devices). If the installation is done on a previously deployed PowerFlex cluster, or used disks, you must tell the installer to overwrite the older data.

    NOTE: If you select this option, the data is erased.
11. Click **Next**.
12. In the **Summary** tab, verify all the configurations and click **Finish**.

13. Click the **Go to Events Page** button to track the installation progress.

Next steps

In case of failure or to check the progress, see the following log files:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform data, see *(Optional) Collect PowerFlex management platform support data*.

Adding new components or features by editing the existing CSV file

Use PowerFlex systems existing CSV topology file to add components or new features to your system.

The CSV file that is used to deploy a system can be used later, in the following scenarios, for adding features or components to an existing PowerFlex system:

- Add storage node
- Add MDMs or tiebreakers. The PowerFlex installer adds them to the MDM cluster and configures them automatically.
- Add SDCs
- Add SDTs
- Configure storage pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles, and so on.

To add and configure components and features, make a copy of your system's latest CSV topology file. Add rows or columns or edit existing rows and columns

Get the latest complete template, see *Related Information*.

The minimum number of nodes for a protection scheme are:

- 2 + 2 is 5 nodes
- 8 + 2 is 11 nodes

NOTE:

- Adding additional storage nodes to the existing protection domain to switch the protection scheme from a 2+2 to 8+2 is not supported.
- A resource group can have multiple protection domain and storage pools with different protection schemes.

The following examples show how you can edit the existing CSV topology file for various modifications.

To add the storage nodes, add a row and update the details

The following example is a 8+2 protection scheme with spare node capacity set as 1. There are 11 nodes.

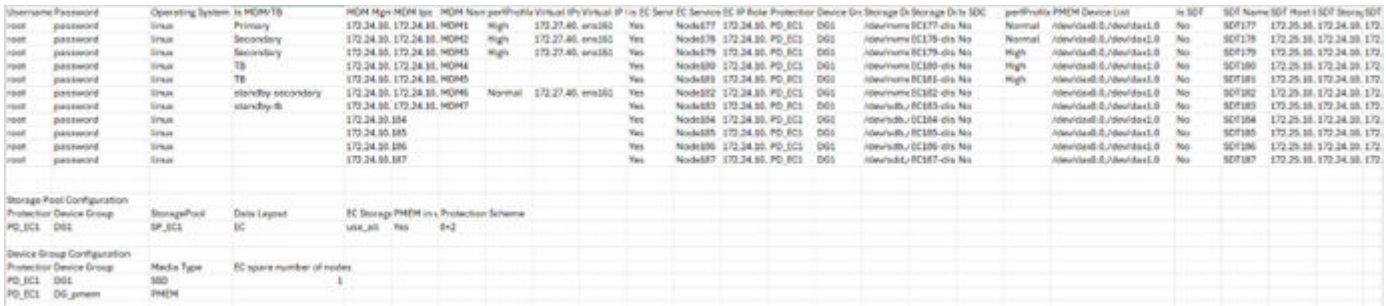


Figure 6. Example CSV file with 8+2 protection scheme

Edited the CSV and new row is added and highlighted in blue.

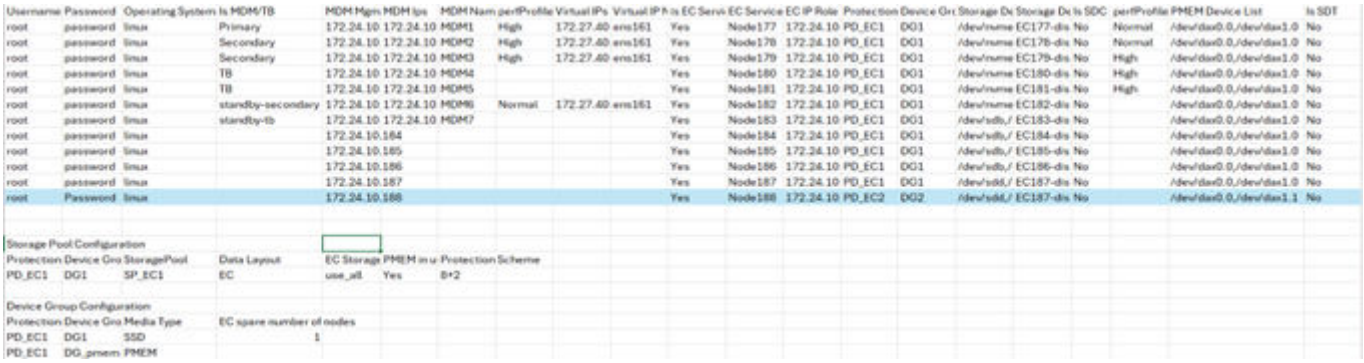


Figure 7. Edited example CSV file with 8+2 protection scheme

The following example is a 2+2 protection scheme with spare node capacity as 1. There are 5 nodes.

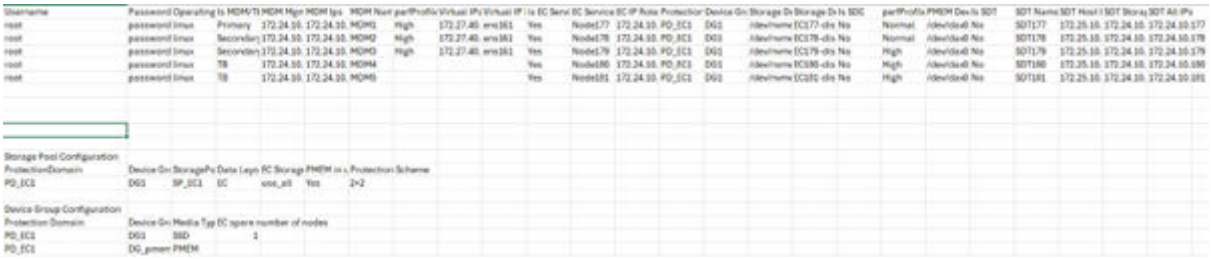


Figure 8. Example CSV file with 2+2 protection scheme

Edited the CSV and new row is added and highlighted in blue.

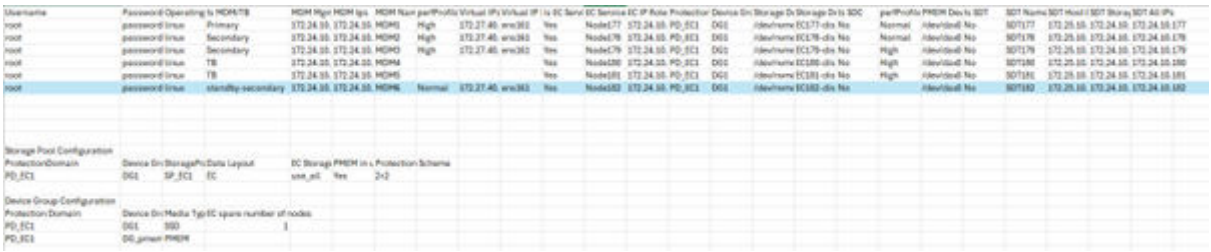


Figure 9. Edited example CSV file with 2+2 protection scheme

In the 2+2 protection scheme, the additional nodes can have the MDM role either as standby secondary or standby tiebreaker or no MDM role at all.

To add storage device to the existing nodes.

Currently, the following CSV example has the devices `/dev/nvme0n1`, `/dev/nvme1n1`, `/dev/nvme2n1`:

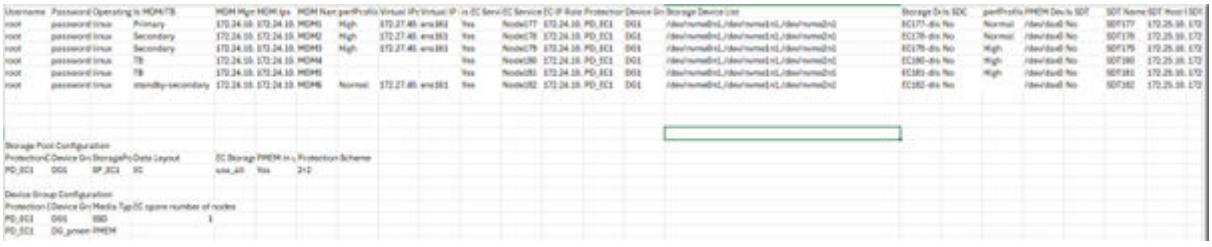


Figure 10. Example CSV file of added storage device to existing node

Edited the CSV with the additional device `/dev/nvme3n1`:

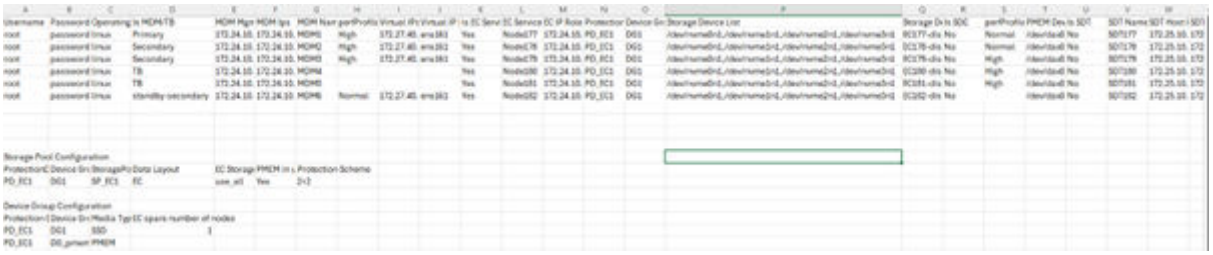


Figure 11. Edited example CSV file of added storage device to existing node

Deploy or expand a cluster using REST API commands

After you have prepared the CSV topology file, use the REST API to parse it and upload the configuration to PowerFlex Manager, and then deploy block storage resources using REST API commands. This procedure is only recommended for advanced users familiar with working with the REST API.

Prerequisites

- Ensure that you have prepared the CSV topology file.
- Ensure that you know the load balancer IP address and user credentials of PowerFlex Manager. All the steps performed in this procedure must be performed using the load balancer IP address. The load balancer IP address is the same IP address you are using to access PowerFlex Manager.
- Ensure that you have the software packages of the PowerFlex core components that you want to deploy, and the ActiveMQ package.
- See the Dell Technologies Developer portal for information about working with the REST API.

About this task

The following procedure provides examples using cURL.

Steps

1. Get the configuration from the CSV file, by parsing it using the PowerFlex REST API. Use the command `POST /im/types/Configuration/instances/actions/parseFromCSV`, where the body is the CSV file reference. For example:

   ```
   curl -v -k -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" -F "file=<PATH_TO_CSV_FILE>" https://<IP_ADDRESS>/im/types/Configuration/
   instances/actions/parseFromCSV
   ```

   Where: *{token}* is the Keycloak token, *<PATH_TO_CSV_FILE>* is the path to the CSV topology file, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.
   The command returns a JSON file containing the required cluster configuration. The installation command requires this file in order to install resources. Save this file.
2. Modify the following with the appropriate values mdmUser, mdmPassword, and liaPassword in the `config.JSON` file. Use the following command to upload the necessary packages needed for deployment:

   ```
   curl -k -X POST -H "Content-Type:multipart/form-data" -H "Authorization: Bearer
   ${token}" --form "files=@<PATH TO ACTIVE MQ RPM>" --form "files=@<PATH TO DGWT
   RPM>" --form "files=@<PATH TO LIA RPM>" --form "files=@<PATH TO MDM RPM>" --form
   "files=@<PATH TO PDS RPM>" --form "files=@<PATH TO SDC RPM>" https://<IP_ADDRESS>/im/
   types/installationPackages/instances/actions/uploadPackages
   ```

3. Verify the packages are uploaded, use the following command:

   ```
   curl -s -k --location -H "Authorization: Bearer ${token}" https://<IP_ADDRESS>/im/
   types/installationPackages/instances
   ```

4. Install block storage resources using the PowerFlex REST API. Use the command `POST /im/types/Configuration/actions/install`, where the body is the configuration that was returned from parsing the CSV file.

If the parameter *systemId* (type string) is added to the configuration, the cluster is created with the given ID. Otherwise, the gateway generates an ID.

Optional parameters:

- noUpload—Skip upload package phase (true or false, default if false)
- noInstall—Skip install package phase (true or false, default if false)
- noConfigure—Skip configure phase and repository files (true or false, default if false)
- noSecurityBootstrap—Skip security bootstrap (part of install phase) (true or false, default if true)
- noLinuxDevValidation (true or false, default if true)
- extend—extend an existing system (true or false, default if false)
- globalZeroPadPolicy (true or false, default if true)
- runAllPhases (true or false, allows to automatically move through all install phases)

For example:

```
curl -v -s -k -i -X POST -H "Content-Type:application/json" -H
"Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Configuration/actions/
install?noSecurityBootstrap=false&runAllPhases=true" -d @<PATH>/config.json
```

Where: *{token}* is the keycloak token, *<IP_ADDRESS>* is the PowerFlex Manager IP address, and *<PATH>* is the path to the JSON configuration file that was output in step 1.

5. Installation is asynchronous and consists of four stages: query, upload, install, and configure. Monitor the progress of the install, use the following command:

```
curl -k -X GET -H "Content-Type: application/json"  -H "Authorization: Bearer $
{token}" "https://<IP_ADDRESS>/im/types/ProcessPhase/actions/queryPhaseState"
```

The installation is complete when the output of phaseStatus is completed and phase stage is configure. The installation takes approximately up to ten minutes to complete.

6. After clearing the gateway, use the following command to set the phase to idle `POSTim/types/ProcessPhase/actions/moveToIdlePhase`:

For example:

```
curl -v -s -k -i -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
${token}" "https://<IP_ADDRESS>im/types/ProcessPhase/actions/moveToIdlePhase"
```

Where `{token}` is the keycloak token, and `<IP_ADDRESS>` is the PowerFlex Manager IP address.

8

Post-installation activities

Perform the procedures listed in the post-installation checklist after system installation to get your PowerFlex storage up and running in an optimal manner.

See the post-installation checklist to determine the procedures that are relevant for your system. Some procedures are performed using PowerFlex Manager, and some are performed using command line. PowerFlex Manager activities are described in detail in the online help system.

Post-installation checklist

The following table lists the configuration tasks that are performed after installing PowerFlex software.

The table also indicates whether tasks are required or optional. It also indicates where you can find documentation for each task.

Table 28. Post-installation configuration tasks

Item	Required or optional?	Follow the procedures in this post-installation section	Follow PowerFlex Manager online help procedures
Enable OpenSSL FIPS compliance	Optional	X	-
Check port usage	Required	X	-
Configure SDC driver for security-enhanced Linux (SELinux)	Optional	X	-
Mount a PowerFlex volume	Optional	X	-
Configure NVMe targets and hosts for block storage if they were not configured by the installation template.	Required	-	X
Add block storage manually, in this order: Add protection domain > Add device group > Add storage node > Add storage pool	Required	-	X
Create and map volumes for block storage if not configured by the installation template	Required	-	X
Associate PowerFlex volumes with physical disks	Optional	X	-
Configure local protection for block storage (local snapshots)	Optional	-	X
Configure local and LDAP users	Required	-	X
Fine tune block storage system performance	Optional	X	-
Configure event and alert notifications to external systems	Optional	-	X
Upload the system license	Required	-	X
Configure SDC access to the MDM	Optional	X	-

Disable kdump on PowerFlex management cluster (KVM) and management virtual machines

Use this procedure to disable kdump on PowerFlex management nodes (KVM).

Prerequisites

NOTE: It is recommended to back up the GRUB file before editing the GRUB file.

Steps

1. SSH to the node and log in as **delladmin**.
2. Verify the service status, type the following: **systemctl status kdump.service**
3. Stop the kdump service, type the following: **sudo systemctl stop kdump.service**
4. Disable the kdump service, type the following: **sudo systemctl disable kdump.service**
5. Modify the GRUB configuration file, type the following:**#vi /etc/default/grub**
   a. Locate the line starting with `GRUB_CMDLINE_LINUX_DEFAULT`.
   b. Change the **crashkernel** parameter to **crashkernel=no**.

      NOTE: If the parameter is **crashkernel=<some_size>**, change it to **crashkernel=no**.

   c. Save the file, press **ESC**+ **:wq**.
6. Update the GRUB configuration file, type the following: **sudo grub2-mkconfig -o /boot/grub2/grub.cfg**
7. Reboot the system, type the following: **sudo reboot**

Automate SDC driver updates due to OS kernel patching

Use the following procedure to deploy SDC-only on SLES, Oracle Linux, or Ubuntu servers.

Ensure that the PowerFlex SDC is compatible with the Linux kernel.

NOTE: All references to Ubuntu in this section apply to supported versions of SLES and Oracle Linux.

To ensure that a client node always has an SDC kernel driver that matches the client kernel, the SDC startup script uses `driver_sync.sh`, a synchronization utility that fetches SDC kernel drivers from a configured repository. This repository can be either the Dell Technologies repository, or a local repository in your private network.

For Linux nodes with PowerFlex SDC drivers:

- The `driver_sync.sh` script is located under `<base_dir>`.
- In the repository, a driver package tarball is located at `<base_dir>/driver_cache/<distribution>/<PowerFlex_version>/<kernel_version>/<driver>.tar`.

Where:

- `<base_dir>` is `/bin/emc/scaleio/<driver>_sync` by default.
- `<distribution>` is one of: CentosStream8, CentosStream9, CoreOS, RH, RHEL8, RHEL9, SLES, SLES15.3, Ubuntu, Ubuntu16.04, Ubuntu18.04, Ubuntu20.04, or Ubuntu22.04
- `<kernel_version>` is the output of `uname -r`.
- `<driver>` is scini for SDC or xcache for RF.

It may be necessary to manually restart the driver/s and services on hosts.

In a normal work flow, an SDC driver is loaded on boot. If there is a mismatch between the version of the running kernel and the kernel against which the driver was compiled, the `driver_sync.sh` script is run automatically to fetch the appropriate driver. If the script fails for any reason (for example, due to incorrect configuration) after fixing the issue with the `driver_sync.sh` script, the driver services must be restarted manually, as described below. Root permission is required to run the commands.

To manually restart driver and services, run this command:

```
service scini restart
```

Ensure that a driver is available for the running kernel

Ensure that the kernel version is correct by performing the following steps.

Steps

1. Ensure that your kernel version matches a version in the repository, by performing the following:

   NOTE: If all servers have the same kernel, you can run this command on a single server.

   a. Run this command on every Ubuntu server:

   ```
   uname -r
   ```

   Output similar to the following is displayed:

   ```
   3.16.0-62-generic
   ```

   b. Copy the following text into a browser, and compare the output to the Ubuntu kernel version in the PowerFlex repository:

   ```
   sftp://mft.dell.com/
   ```

   NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. To use a mirror repository, you must ensure that the SSH public and SSH private keys exist in all system nodes in the same path.

   The private key should have directory-level permission only (chmod 600 <private_key_path>).

3. The GPG key must be located in all system nodes in the same path. This key is required for using a mirror or Dell Technologies repository.

Next steps

The configuration of `driver_sync.sh` is specified in `driver_sync.conf` file, which can be created using the following method:

- Create the configuration file manually

NOTE: The SDC package now supports automated installation on Linux operating systems with SELinux. For supported operating systems, see the support matrix. To ensure that SELinux is activated and that SDCs are deployed as expected, verify that the relevant pre-requisite packages are installed.

Parameters to add to the driver_sync file

Add the following parameters to the driver synchronization utility.

Steps

1. Validate that the repository contains the kernel your operating system requires.

   ```
   sftp://mft.dell.com/
   ```

   Or

   ```
   https://mft.dell.com/
   ```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download `sftp_keys_pkb.tar` from `<location needed>` and extract the contents to your SDC/RF host, as user root.
3. Place scini_repo_key.pub at the location specified in repo_public_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_repo_key.pub` by default.
4. Place scini_key at the location specified in user_private_rsa_key, `/etc/emc/scaleio/<driver>_sync/scini_key` by default. Where `<driver>` is scini for the SDC.
5. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

6. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
repo_user = PowerFlexRepo
user_private_rsa_key = /etc/emc/scaleio/scini_sync/scini_key
repo_public_rsa_key = /etc/emc/scaleio/scini_sync/scini_repo_key.pub
```

NOTE: The `repo_password` line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

7. Restart the Storage Data Client RFCache driver:
   - Option1: Run from the operating system command prompt:

     ```
     systemctl restart <driver>
     ```

     NOTE: You can restart the Storage Data Client from the operating system only if the operating system is not using any volumes.
   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
8. Check that the `<driver>` is running:

```
systemctl status <driver>
```

9. Once you are finished, perform an operating system update and reboot to load the new kernel. The Storage Data Client driver sync automatically downloads the kernel object (KO) file, if available.

Create the configuration file manually

The `driver_sync.conf` file can be created or edited manually using a text editor. This topic is relevant for deployments on Linux flavors other than RedHat.

About this task

The file exists in the `etc` folder, with dummy values, and you must edit that file, and then copy it to the `etc` folder on every SDC server in the system. The `etc` folder is located in:

- SDC

Table 29. SDC host and path

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

The configuration of the `driver_sync` script is read from `driver_sync.conf` (an example appears below), which contains the following parameters:

Table 30. driver_sync.conf parameters

Parameter	Description
repo_address	Encodes the chosen method and address.
repo_user	Identifies the username that is used when logging in to the drivers server. Required for SFTP and FTP methods.
repo_password	The FTP server login password (for user repo_user). Required for FTP method
user_private_rsa_key	Contains the path of the private RSA key file that is used to SFTP connection. This key file (provided by PowerFlex on installation) enables SFTP connection without a password.
local_dir	The local directory in which the downloaded SDC/RF drivers are stored. For SDC/RF clients, this value is set up during installation and does not require a change.
repo_public_rsa_key	The path of the public RSA key of the repository server machine (also known as the host public key).
module_sigcheck	Determines whether to verify (1) or not to verify (0) the downloaded kernel drivers.
emc_public_gpg_key	A key that is used to verify signatures of downloaded drivers. Enter the path to the public GPG signing key of PowerFlex (like the one that comes with PowerFlex LIA). This is needed only if module_sigcheck is set to 1.
The following parameters are used when driver_sync.sh is used to synchronize a local repository against a remote one.	
sync_pattern	A regular expression that can be used to fetch several drivers at once. This parameter is not needed, and becomes relevant only when your own private repository uses driver_sync.sh to fetch from a remote (For example: Dell) repository.
is_local_repo	Set to 0 (default) or omit, for SDC/RF clients. Set to 1 when driver_sync.sh is used to synchronize local repository against a remote repository. Setting to 1 causes the fetched tar files not to be extracted.

For example:

```
################################################
#driver_sync Configuration file
#Everything after a '#' until the end of the line is ignored
################################################
#Repository address, prefixed by protocol
repo_address = sftp://localhost/path/to/repo/dir
#repo_address = ftp://localhost/path/to/repo/dir
#repo_address = file://local/path/to/repo/dir
# Repository user (valid for ftp/sftp protocol)
repo_user = scini
# Repository password (valid for ftp protocol)
repo_password = scini
# Local directory for modules
local_dir = /bin/emc/scaleio/scini_cache/
# User's RSA private key file (sftp protocol)
user_private_rsa_key = /bin/emc/scaleio/scini_key
# Repository host public key (sftp protocol)
repo_public_rsa_key = /bin/emc/scaleio/scini_repo_key.pub
# Should the fetched modules' signatures be checked [0, 1]
module_sigcheck = 1
# EMC public signature key (needed when module_sigcheck is 1)
emc_public_gpg_key = /bin/emc/scaleio/emc_key.pub
# Sync pattern (regular expression) for massive retrieve
sync_pattern = .*
```

Perform the following steps for updating the driver_sync file on the storage data client:

Steps

1. Validate that the SFTP contains the kernel your operating system requires.

```
sftp://mft.dell.com/
```

Or

```
https://mft.dell.com/
```

NOTE: You can use Hypertext Transfer Protocol Secure (HTTPS) while checking driver availability manually.

2. Download and extract the attached sftp_keys to your SDC host, as user root, to /etc/emc/scaleio/scini_sync. Find the file that is attached to the article: sftp_keys_pkb.tar.
3. Change the permissions of a private key file.

```
chmod 600 /etc/emc/scaleio/scini_sync/scini_key
```

4. Edit /etc/emc/scaleio/scini_sync/driver_sync.confsimilar to the example below:

```
repo_address = sftp://mft.dell.com
                                    repo_user = PowerFlexRepo
                                    user_private_rsa_key = /etc/emc/scaleio/
scini_sync/scini_key
                                    repo_public_rsa_key = /etc/emc/scaleio/scini_sync/
scini_repo_key.pub
```

NOTE: The repo_password line is not used with SFTP and can be left with the default value. Use the real password supplied above to check driver availability.

5. Restart the storage data client driver:
   - Option1: Run from the operating system command prompt:

     ```
     systemctl restart scini
     ```

     NOTE: This is possible only if no volumes are in use by the operating system.
   - Option2: If needed, plan for a reboot cycle of the operating system for this setting to take effect.
6. Check that the `sciniservice` is running:

```
systemctl status scini
```

7. Once you are finished, perform an operating system update and reboot to load the new kernel. The storage data client driver sync automatically downloads the kernel object (KO) file, if available.

Update the PowerFlex signature key

If the PowerFlex package-signing key expires, you must update the configuration for each SDC.

About this task

When a PowerFlex package-signing key expires, a new public key is issued. You must update `driver_sync.conf` to use the new public key.

Steps

1. Download the new public key from the Dell Technologies Support site.
2. On each SDC node, edit `driver_sync.conf` so that the value of `emc_public_gpg_key` is the path to the new key file.

   The expired public key will still be in the gpg keyring, and older packages that were signed with the old key can still be verified.

   Location of `driver_sync.conf`:

- SDC

Table 31. SDC host and path

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

Create a mirror repository

For Ubuntu or Oracle Linux (OL, also known as OEL), set up a local mirror repository (optional).

This section describes how to set up a local repository. This is necessary for scenarios when connecting all SDC nodes to the PowerFlex repository is not possible, perhaps due to security considerations or other limitations.

A common workflow would be to configure your mirror repository to synchronize with the PowerFlex repository, while your SDC nodes will fetch drivers (synchronize) from the mirror repository.

Installing the SDC

If a host of the SDC type was not installed on front-end servers during initial installation activities, follow the appropriate instructions for your front-end servers' operating system.

See the *Dell PowerFlex 5.0.x Support Matrix* to ensure that the intended SDC operating system of the host is compatible for use with your PowerFlex system.

You can use any of the following methods to install the SDC:

Table 32. SDC installation methods

Operating system	Methods
VMware ESXi	• PowerFlex Manager procedure for adding resources to an existing system. • Command line (esxcli)
Linux	• PowerFlex Manager procedure for adding resources to an existing system (not applicable for Linux). • Command line
AIX	Command line
Windows	• PowerFlex Manager procedure for adding resources to an existing system. • Command line
Citrix XEN(XenServer)	• PowerFlex Manager procedure for adding resources to an existing system.

Install the SDC on an VMware ESXi server

Install SDC on VMware ESXi servers using command-line esxcli commands.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.

- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site) .
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires two servers reboots.

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:
   a. Change the directory where you have copied SDC.ZIP file.
   b. Unzip the SDC.ZIP file, run the following command: **`unzip <filename.zip>`**.
   c. Install the SDC package, run the following command: **`esxcli software vib install -v <path_to_vib>`**.

   Where **`<path_to_vib>`** is the path to SDC.VIB file which is unzipped in the above command.

3. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.

4. Set the IP address of the MDM:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced settings in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

```
/bin/auto-backup.sh
```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

Output similar to the following indicates that the driver is loaded:

```
scini 2 808
```

The SDC component is installed on the VMware ESXi-based server.

Results

The SDC is installed on the VMware ESXi-based server and is connected to PowerFlex.

Install the SDC on a Linux-based server

Install the storage data client on a Linux-based server with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more storage data clients to an existing system, and for adding storage data clients to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Citrix Hypervisor(XenServer)/Red Hat Enterprise Linux/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Install the SDC on an AIX-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server.

Steps

1. From the command line, install the SDC:

   ```
   MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this: `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`

2. Optional: For SDC authentication to the MDM for access, enter the following command:

   ```
   chdev -l scini -a "mdm_password=<ip>-<password>"
   ```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Install SDC on Windows-based PowerFlex compute-only node

Use this procedure to install SDC on the Windows-based PowerFlex compute-only node.

Steps

1. Download PowerFlex LIA and the SDC plugin.
2. Double-click the PowerFlex LIA setup file.
3. Accept the terms and license agreement and click **Install**.
4. Click **Finish**.
5. Do the following for Windows-based SDC node before deploying the SDC component using the command prompt with administrator privileges:
   a. Type **`bcdedit /debug ON`**
   b. Type **`bcdedit /`**set testsigning ON
   c. Type **`reboot windows`**

6. If you know the MDMs VIPs before installing the SDC component, run the command:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=<LIST_VIP_MDM_IPS>
```

   For example:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=192.168.151.185,192.168.152.185
```

7. Accept the terms and license agreement and click **Install**.
8. Click **Finish**.
9. On confirmation from the customer to reboot the Windows server, perform a reboot to load the SDC driver on the server.
10. If the MDMs VIPs are not provided before installing the SDC component:
    a. Double-click the PowerFlex SDC setup.
    b. Accept the terms in the license agreement and click **Install**.
    c. Click **Finish**.
    d. Type **`c:\Program Files\EMC\scaleio\sdc\bin>drv_cfg.exe --add_mdm --ip <VIPs_MDMs>`** to configure the PowerFlex node in the PowerFlex software.

Mapping volumes to Windows server

Use this procedure to map volumes to the Windows server.

Steps

1. Log in to PowerFlex Manager.
2. Verify the Windows PowerFlex compute-only nodes listed under **Lifecysle** > **Storage nodes**.
3. Click **Block** > **Volumes** .
4. Select the **Volume** which needs to mapped to.
5. Click **Provision** > **MAP**.
6. Select the newly add Windows host and click **MAP**.
7. Log in to the Windows PowerFlex compute-only nodes and select **Disk Management**.
8. Rescan the disk and bring it online.
9. Click **Initialize disk**.
10. Create **New Simple Volume** and click **Next**.
11. Select **Default** and click **Next**.
12. Assign the drive letter.
13. Select **Default** and click **Next**.
14. Click **Finish**.

Enable OpenSSL FIPS compliance on the storage node, MDM, LIA

Enable the implementation of OpenSSL Federal Information Processing Standards (FIPS) compliance on the storage node, MDM, LIA for communication between the external components, including PowerFlex Manager, PowerFlex gateway, and CLI, to the MDM. It is also enabled for any other usage of the OpenSSL library.

Prerequisites

The MDM must be hosted on Linux with the OpenSSL package installed.

Steps

1. On each host running PowerFlex, open the configuration file of each component with a text editor.
   The configuration file is `/opt/emc/scaleio/<COMPONENT>/cfg/conf.txt`, where *<COMPONENT>* is the lowercase name of the component (for example, pds).
2. Add the parameter `security_enable_fips=1` to the file.
3. Save and close the file.
4. Open the SCLI configuration file with a text editor:
   The configuration file is at: `~/.scli/conf.txt`.
5. Add the parameter `security_enable_fips=1` to the file.
6. Save and close the file.
7. On each host, restart each component's service:

```
service pds restart
service dgwt restart
```

8. Verify that OpenSSL FIPS compliance has been enabled by running:

```
cat  /proc/sys/crypto/fips_enabled
```

   If it has been enabled correctly, the output should be `1`. If the output is not `1`, enable OpenSSL FIPS at the operating system level.

Check port usage

If you are using a firewall to protect your network, and need to enable ports and IP addresses, see the *Dell PowerFlex Family 5.x Security Configuration Guide*. This guide contains information about port usage for all components, and what to modify in case port numbers need to be changed.

Configure the SDC-only (scini) driver for SELinux

SELinux may prevent the scini driver from starting on Red Hat Enterprise Linux-based SDC nodes. This procedure prevents this issue.

About this task

If SELinux is enabled and enforced, it can prevent the SDC-only driver (scini) from starting on Red Hat Enterprise Linux-based nodes.

NOTE: To remove the SELinux module, run: `semodule -r scini`

Steps

1. On the SDC, use a text editor to create two text files in any directory:
   a. Create a file called `scini.fe`, and add the following text:

```
/usr/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
```

   b. Create a file called `scini.te`, and add the following text:

```
module scini 1.0;

require {
        type insmod_t;
        type modules_object_t;
        class system module_load;
}
```

2. Build and install the SELinux module, by running the following commands:

```
checkmodule -M -m scini.te -o scini.mod
semodule_package -o scini.pp -m scini.mod -f scini.fe
semodule -i scini.pp
restorecon -R -v /bin/emc/scaleio/
```

3. To check the current configuration, run the command:

```
sestatus
```

Output similar to the following should be displayed, showing that *SELinux status* is enabled, and *Current mode* is enforcing:

```
SELinus status: enabled
SELinuxfs mount: /sys/fs/selinux
SELinux root directory: /etc/selinux
Loaded policy name: targeted
Current mode: enforcing
Mode from config file: enforcing
Policy MLS status: enabled
Policy deny_unknown status: allowed
Memory protection checking: actual (secure)
Max kernel policy version: 31
```

4. To check that the scini module is configured in SELinux, run the command:

```
semodule --list=full | grep scini
```

Output similar to the following should be displayed:

```
400    scini                pp
```

5. Start the scini service:

```
service scini start
```

Mount a PowerFlex volume on SDC-only

The exposed PowerFlex volumes are connected to the servers by the network. To configure the mounting options of PowerFlex devices, follow the instructions for your specific Linux-based operating system, below.

About this task

Use persistent device names since the `/dev/sciniX` names are not guaranteed to persist between reboots.

Steps

1. Determine the `/dev/disk/by-id` correlation to `/dev/sciniX`:

```
ls -l /dev/disk/by-id/ |grep scini
```

Output similar to the following appears:

```
lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f0330000000a ->
../../scinia
lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f03200000009 ->
../../scinib
lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f02c00000003 ->
../../scinic
```

2. Run the mount command:

```
/dev/disk/by-id/<EMC-vol-id> <mount point>
```

Example:

```
mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia
```

3. To make the mount command persistent, edit the `/etc/fstab` file according to the instructions for your operating system:
   - Red Hat Enterprise Linux 7.x:

     In `/etc/fstab`, use a text editor to add `_netdev` to the PowerFlex mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs
     defaults,_netdev 0 0
     ```

     Ensure that you comply with the `netdev` and syntax rules for your file system, as described in the `man` page.

   - SLES:

     In `/etc/fstab`, use a text editor to add `nofail` to the mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs nofail 0 0
     ```

     Ensure that you comply with the `nofail` and syntax rules for your file system, as described in the `man` page.

Associate PowerFlex volumes with physical disks on MDM-only

If necessary, you can associate PowerFlex volumes with physical disks.

To get PowerFlex volume information, run the CLI command `scli --query_all_volumes` (or `--query_all` or `--query_volume`).

Output similar to the following appears:

```
Query-all-volumes returned 10 volumes
Protection Domain 0728185d00000000 Name: pd1
Storage Pool ad99eaab00000000 Name: default
<No volumes defined>

Storage Pool ad99eaac00000001 Name: sp1
Volume ID: fac22a6300000000 Name: vol0 Size: 152.0 GB (155648 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6400000001 Name: vol1 Size: 400.0 GB (409600 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6500000002 Name: vol2 Size: 80.0 GB (81920 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6600000003 Name: vol3 Size: 392.0 GB (401408 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6700000004 Name: vol4 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6800000005 Name: vol5 Size: 112.0 GB (114688 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6900000006 Name: vol6 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6a00000007 Name: vol7 Size: 176.0 GB (180224 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6b00000008 Name: vol8 Size: 272.0 GB (278528 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6c00000009 Name: vol9 Size: 360.0 GB (368640 MB) Mapped to 1 SDC Thin-provisioned
```

This output shows the volume ID and name, and other volume information.

Retrieve volume information for Linux from SDC-only

Retrieve volume information from the SDC host.

On the SDC host, run the following command to get the operating system volume information that correlates to the PowerFlex scini device name:

```
ls -l /dev/disk/by-id/ |grep scini
```

Output similar to the following appears:

```
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6300000000 -> ../../scinia
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6400000001 -> ../../scinic
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6500000002 -> ../../scinib
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6600000003 -> ../../scinie
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6700000004 -> ../../scinid
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6800000005 -> ../../scinif
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6900000006 -> ../../scinig
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6a00000007 -> ../../scinii
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6b00000008 -> ../../scinih
lrwxrwxrwx 1 root root 12 Aug 25 19:43 emc-vol-62c093a52d14aec7-fac22a6c00000009 -> ../../scinij
```

This output shows the scini volume name and the volume ID.

By matching the volume ID in both outputs, you can match the operating system name, sciniX, with the PowerFlex volume name.

For example:

- scinia = fac22a6300000000 = vol0
- scinic = fac22a6400000001 = vol1

Alternatively, run the `sg_inq /dev/sciniX` SCSI query command (which requires that the sg3_utils be installed on the Linux host). The result of this command includes the EMC volume ID at the bottom of the output, as illustrated:

```
Vendor identification: EMC
Product identification: ScaleIO
Product revision level: 1.3
Unit serial number: EMC-62c093a52d14aec7-fac22a6300000000
```

NOTE: The product identification remains as ScaleIO (not PowerFlex).

Add PowerFlex volumes to Linux LVM physical volumes on SDC-only

Add PowerFlex volumes to Linux LVM physical volumes.

Steps

1. In the `/etc/lvm/lvm.conf` file add the following lines:

```
types = [ "scini", 16 ]
```

2. After configured, the `lvmdiskscan` command should yield results similar to the following:

```
/dev/scinia  [      96.00 GiB] LVM physical volume
/dev/scinib  [     320.00 GiB] LVM physical volume
/dev/scinic1 [      56.00 GiB]
/dev/scinid  [      32.00 GiB]
1 disk
1 partition
2 LVM physical volume whole disks
0 LVM physical volumes
```

3. Continue with normal LVM steps.

Retrieve volume information for Windows from MDM-only

You can retrieve volume information from the MDM.

About this task

The `sg_inq.exe` file was added to the MSI installation and can be found at `C:\Program Files\EMC\ScaleIO\SDC\diag\`.

Steps

1. Run the `sg_inq HardiskX` SCSI query command.
   The result of this command includes the EMC volume ID at the bottom of the output.
2. On the MDM, get the PowerFlex volume information:

   ```
   C:\Program Files\emc\scaleio\sdc\bin\drv_cfg --query_vol
   ```

   Output similar to the following is displayed:

   ```
   Retrieved 5 volume(s)
   VOL-ID 6acb988100000000 MDM-ID 0b246c9a755ca3dd
   VOL-ID 6acb988200000001 MDM-ID 0b246c9a755ca3dd
   VOL-ID 6acb988300000002 MDM-ID 0b246c9a755ca3dd
   VOL-ID 6acb988400000003 MDM-ID 0b246c9a755ca3dd
   VOL-ID 6acb988500000004 MDM-ID 0b246c9a755ca3dd
   ```

3. From the Windows command prompt, run this command:

   ```
   wmic diskdrive get deviceid,serialnumber | findstr "EMC"
   ```

   Output similar to the following is displayed:

   ```
   \\.\PHYSICALDRIVE13 EMC-0b246c9a755ca3dd-6acb988500000004
   ```

   The first part of the output is the disk name. In our example: `PHYSICALDRIVE13`

   The second part is the disk serial number. The last set of the second part (after the dash) is the PowerFlex volume ID. In our example: `6acb988500000004`

Retrieve volume information for AIX from SDC-only

On AIX servers, associate the PowerFlex volume ID with the AIX physical device.

Steps

1. On the SDC host, run the following command to get the operating system volume information:

   ```
   #odmget -q "name like scinid* and attribute=vol_id" CuAt
   ```

   Output similar to the following is displayed:

   ```
   CuAt:
   name = "scinid0"
   attribute = "vol_id"
   value = "e120a92d00000000"
   type = "R"
   generic = "D"
   rep = "s"
   nls_index = 22
   [root@cnode02 /]#odmget -q "name like scinid* and attribute=vol_id" CuAt
   ```

```
CuAt:
name = "scinid2"
attribute = "vol_id"
value = "e120a92f00000002"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid8"
attribute = "vol_id"
value = "e120a93500000008"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid0"
attribute = "vol_id"
value = "e120a92d00000000"
type = "R"
generic = "D"
rep = "s"
nls_index = 22
```

You can get information for a single volume by using this command:

```
#odmget -q "name=scinid0 and attribute=vol_id" CuAt
```

2. Match the value of the `value` field with the PowerFlex volume ID.

Performance tuning post-installation

This section describes steps to take after completing a successful installation to enhance performance.

You can improve the PowerFlex system performance in terms of IOPS, latency, and bandwidth, by performing environment-specific fine-tuning on the operating system, network, and PowerFlex components.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, it is highly recommended to thoroughly test all changes. For further assistance, contact Dell Technologies Support.

PowerFlex system changes for PowerFlex components using MDM

This section describes the various system changes available for enhancing PowerFlex performance.

Using the set_performance_parameters utility for MDM and storage nodes

The following table describes the commands for specific tasks:

Table 33. set_performance_parameters commands

Action	Command
View current settings.	`scli --query_performance_parameters`

Table 33. set_performance_parameters commands (continued)

Action	Command
By default is High_performance. Run the command to change to High_performance if the value is not set to High_performance.	`scli --set_performance_parameters --all_storage_node --all_sdc --apply_to_mdm --profile high_performance`
Change the profile back to compact.	`scli --set_performance_parameters --all_storage_node --all_sdc --apply_to_mdm --profile compact`
View full parameter settings for an MDM.	`scli --query_performance_parameters --print_all`
View the full parameter settings of a specific storage node (this also shows the MDM settings).	`scli --query_performance_parameters --storage_node_name<NAME> --print_all`
View the full parameter settings of a specific SDC (this also shows the MDM settings).	`scli --query_performance_parameters --sdc_name<NAME> --print_all`

Jumbo frames and the potential impact on performance for storage nodes and SDC-only

Enabling jumbo frames can result in approximately 10% improvement in performance if all network components fully support jumbo frames. If some network components do not fully support jumbo frames, it is recommended to use the default setting; MTU 1,500.

Before activating MTU settings on the logical level, set jumbo frames = MTU 9000 on the physical switch ports that are connected to the node. Failure to do so may lead to network disconnects and packet drops.

See your relevant vendor guidelines on how to configure jumbo frame support.

Configure jumbo frames for Linux

Configure jumbo frames for NIC cards in the Linux-based PowerFlex nodes.

Steps

1. Run the `ifconfig` command to get the NIC information.
2. Depending on the operating system, run the command:

Operating system	Command
Red Hat Enterprise Linux	Edit `/etc/sysconfig/network-scripts/ifcfg-` *<NIC_NAME>*
SLES	Edit `/etc/sysconfig/network/ifcfg-`*<NIC_NAME>*

3. Add the parameter `MTU=9000` to the file.
4. To apply the changes, type: **`service network restart`**
5. Run `ifconfig` again to verify that the settings have been changed.
6. To test the command, type: **`ping -M do -s 8972`***<DESTINATION_IP_ADDRESS>*
   Output should look similar to the following test ping to ensure MTU setting:

```
[root@116BT-18 network-scripts]# ping -M do -s 8972 10.108.158.48
PING 10.108.158.48 (10.108.158.48) 8972(9000) bytes of data.
8980 bytes from 10.108.158.48: icmp_seq=1 ttl=64 time=0.031 ms
8980 bytes from 10.108.158.48: icmp_seq=2 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=3 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=4 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=5 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=6 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=7 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=8 ttl=64 time=0.029 ms
8980 bytes from 10.108.158.48: icmp_seq=9 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=10 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=11 ttl=64 time=0.012 ms
8980 bytes from 10.108.158.48: icmp_seq=12 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=13 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=14 ttl=64 time=0.027 ms
^C
--- 10.108.158.48 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13581ms
rtt min/avg/max/mdev = 0.012/0.025/0.031/0.007 ms
```

Figure 12. Test ping of MTU setting

Configure jumbo frames for Windows

Configure jumbo frames for NIC cards in the Windows-based PowerFlex nodes.

Steps

1. Change the `Maximum Transmission Unit (mtu)` setting to `9,000`, or the highest value that is supported by the switch and the connected nodes.
2. Determine the appropriate NIC name by typing the command:

   ```
   netsh interface ipv4 show interface
   ```

   Sample output:

   ```
   Idx     Met         MTU          State                Name
   ---  ----------  ----------  ------------  ---------------------------
     1          75  4294967295  connected     Loopback Pseudo-Interface 1
    10           5        1500  disconnected  Ethernet
     3          25        1500  connected     mgmt
     6           5        1500  disconnected  Ethernet 3
    12           5        1500  disconnected  Ethernet 4
     7          15        1500  connected     Data1
     9          15        9028  connected     Data2
   ```

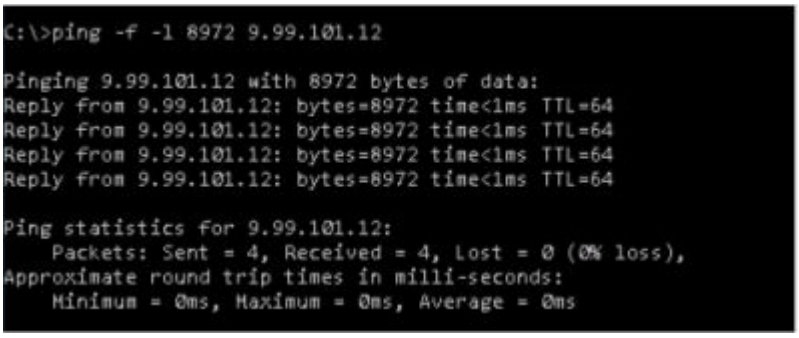
   In this example, index 7 is the appropriate network.
3. Type the command:

   ```
   netsh interface ipv4 set subinterface <network_ID> mtu=9000 store=persistent
   ```

   Where *network_ID* is the ID from the output in the previous step; in this case, the ID is 17.
4. In the **Advanced** tab of the **Adapter Properties** dialog box for your vendor and driver, change the value of `Jumbo Packet` to *9000*.
5. Click **OK**.
   The network connection may disconnect briefly during this phase.
6. Verify that the configuration is working, by typing the command:

   ```
   ping -f -l 8972 <Destination_IP_Address>
   ```

   Output similar to the following should display:



NOTE: Ensure that the switch supports 10 GB Ethernet.

Configure jumbo frames for VMware ESXi

Configure jumbo frames for NIC cards in the VMware ESXi-based PowerFlex nodes.

Steps

Change the `Maximum Transmission Unit (mtu)` setting to `9,000` on the vSwitches and on the SVM (be sure to make the change in `/etc/sysconfig/network/ifcfg-ethX`):

a. Type the command:

```
esxcfg-vswitch -m 9000 <vSwitch>
```

b. Create a VMkernel with jumbo frames support by typing the following commands:

i.
```
esxcfg-vswitch -d
```

ii.
```
esxcfg-vswitch -A vmkernel# vSwitch#
```

iii.
```
esxcfg-vmknic -a -i <ip address> -n <netmask> -m 9000 <portgroup name>
```

NOTE: Changing jumbo frames on a vSwitch will not change the VMkernel MTU size. For older vCenter versions, check to ensure the MTU setting has been changed. If it was not successful, users may need to delete and re-create the VMkernel. For newer vCenter versions, modify the MTU of VMkernel using the vSphere web client.

Optimize Linux on storage nodes

When using the SSD devices, Dell Technologies recommends that the I/O scheduler of the devices be modified.

1. Change the disk scheduler in grub by appending "elevator=noop" to the variable `GRUB_CMDLINE_LINUX_DEFAULT`.

   For example, in `/etc/default/grub`, `GRUB_CMDLINE_LINUX_DEFAULT="quiet splash elevator=noop"`

2. Recompile the grub.cfg file

   For example, `grub2-mkconfig -o /boot/grub2/grub.cfg`

3. Reboot the Linux system.

NOTE: To make these changes persistent after reboot, either create a script that runs on boot, or change the kernel default scheduler using kernel command line.

Change the GRUB template for CPUs on storage nodes and SDC nodes

For CPU throughput on PowerFlex nodes running Red Hat Enterprise Linux 7.2 or later, change the GRUB template on every node to ensure that the CPUs maintain good latency performance.

Steps

1. Open the GRUB template for editing:

```
vim /etc/default/grub
```

2. Find the `GRUB_CMDLINE_LINUX` configuration option and append the following to the line:

```
intel_idle.max_cstate=1 intel_pstate=disable
```

   For example:

```
GRUB_CMDLINE_LINUX="crashkernel=auto rd.lvm.lv=rhel/root
rd.lvm.lv=rhel/swap rhgb intel_idle.max_cstate=1
intel_pstate=disable quiet"
```

3. Compile the new GRUB:

```
grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

4. Stop and then disable `tuned`:

```
systemctl stop tuned
systemctl disable tuned
```

5. Reboot the node:

```
reboot
```

6. Ping the node to ensure that the GRUB change was implemented.
   The ping time should be in the 0.03 ms range.
7. Repeat the procedure on every relevant node.

Optimize VMware ESXi on storage nodes and SDC nodes

To improve I/O concurrency, you can increase the per device queue length value on a per data store basis. Per device queue length is seen as `No of outstanding IOs with competing worlds` in the display output.

Use the following command to increase the queue length:

```
esxcli storage core device set -d <DEVICE_ID> -O <Outstanding IOs>
```

Where *<Outstanding IOs>* can be a number ranging from 32–16384 (default=32). Dell Technologies recommends increasing queue length to 256.

Example:

```
esxcli storage core device set -d eui.16bb852c56d3b93e3888003b00000000 -O 256
```

Optimize Windows on storage nodes and SDC

The delayed acknowledgment feature causes high latencies for low I/O rates. Dell Technologies recommends that you disable the delayed ack feature on every network interface.

To change each SDC interface, excluding the management IP address, add the following registry entries:

- `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\<SAN interface GUID>`
- Entries: **`TcpAckFrequency TcpNoDelay`**
- Value type: **`REG_DWORD`**, number
- Value to disable: **`1`**

NOTE: Depending on the Windows version, the method for changing delayed ack varies. See your relevant vendor guidelines for specific details.

Ensure that the `customize power plan` is set to high performance, as shown in the following figure:

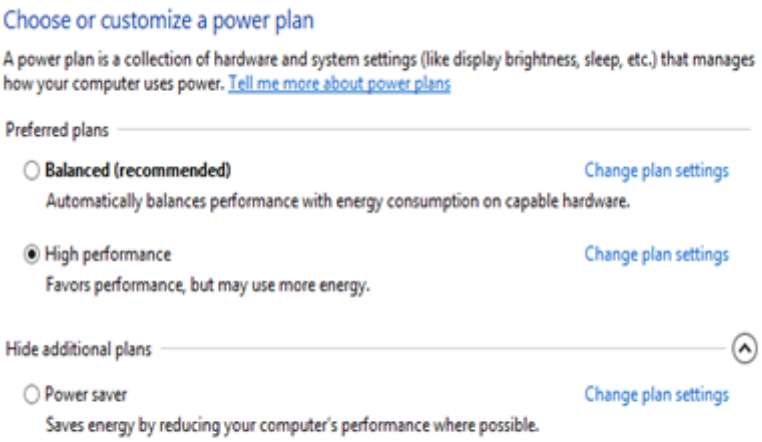


Figure 13. Customize power plan

Optimize Hyper-V

To improve Hyper-V performance and to minimize unnecessary resource consumption, review the following requirements:

- Disable NUMA spanning on Hyper-V hosts.
- Install SDC on the root partition.
- Dell Technologies recommends that you do not create virtual switches on data networks. Enable jumbo frames on virtual switches.

Optimize AIX

The following options optimize the AIX operating system:

- To turn off kernel tracker, run the following command:

```
bosedbug
```

- To reboot an AIX node, run the following command:

```
shutdown -r
```

- To review settings as viewed in the table, run the following command:

```
bosdebug
```

Table 34. AIX settings

AIX settings	Description
Memory debugger	off
Memory sizes	0
Network memory sizes	0
Kernel debugger	off
Real-time kernel	off
Backtracking fault log	off
Kernext memory tracking	off

Configure SDC access to the MDM

It is possible to restrict SDC access to the MDM, pending approval of the SDC by the system. The default system setting is full access (restricted SDC mode is disabled).

When the restricted SDC mode is enabled, volumes can only be mapped to approved SDCs. Set a restricted SDC mode using the CLI. Approval is obtained by issuing the `--add_sdc` command for each SDC. You can set a restricted mode before or after SDCs have been added to your network. You can also approve SDCs using PowerFlex Manager.

The following CLI commands manage this feature:

Table 35. CLI commands

Action	Command
Enable or disable restricted SDC mode.	`set_restricted_sdc_mode command`
Add an SDC to the approved list, when restricted SDC mode is enabled.	`--add_sdc`

For more information, see the *Dell PowerFlex 5.0.x CLI Reference Guide*.

9

Optional installation activities

This section contains miscellaneous installation activities that may not be required for the installation.

Connecting hosts to a PowerFlex system

Configuring NVMe initiators as hosts connecting to PowerFlex system

Use the following procedures to configure the NVMe initiators on hosts for Linux and VMware ESXi-based systems.

The procedure may vary depending on the operating system versions. For more information, see the operating system vendor documentation.

NOTE: NVMe over TCP initiator host-side support is only available for VMware ESXi and SUSE. All other Linux platforms are considered as technology preview.

Configure NVMe initiators on hosts for Linux-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example only. Consult your operating system's documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

Prerequisites

Ensure that you have the login credentials for each host.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running Linux operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 5.0.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Ensure that the NVMe modules are loaded on the NVMe initiator:
   a. Log in to the host.
   b. In the command line, run the command:

      ```
      lsmod | grep nvme
      ```

   c. Ensure that `nvme_tcp` and `nvme_fabrics` modules are loaded.
      If they are loaded, go to step 2. If they are not loaded, continue to the next substep.
   d. Perform one of the following procedures:

- For manual loading of modules (perform after every reboot), run:

```
modprobe nvme_tcp
modprobe nvme_fabrics
```

- For automatic loading of NVMe modules, run:

```
echo nvme_tcp >> /etc/modules-load.d/nvme_tcp.conf
echo nvme_fabrics >> /etc/modules-load.d/nvme_tcp.conf
```

2. Ensure that the NVMe CLI is installed on the initiator:
   a. Run the following command:

```
nvme
```

   If the CLI responds, go to step 3. If it does not, go to the next substep.
   b. Run the command:

```
dnf install nvme-cli
```

3. Ensure that a unique host NQN and unique host ID are configured in the initiator.
   a. To verify that a host NQN is present, run the command:

```
nvme show-hostnqn
```

   Example of a host NQN:

```
nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0051-5610-8046-b5c04f484b32
```

   b. If no host NQN exists, generate one by running the command:

```
nvme gen-hostnqn > /etc/nvme/hostnqn
```

   NOTE: Make a note of each PowerFlex node's host NQN. You need this information when configuring the NVMe targets with PowerFlex Manager.

   c. To verify that a host ID is present, run the command:

```
cat /etc/nvme/hostid
```

   Example of a host ID:

```
4c4c4544-0051-5610-8046-b5c04f484b32
```

   Last 36 characters of a nvme gen-hostqn name can be used. The format must be precise, otherwise the connection fails.
4. Perform the following actions for each required protection domain:
   a. Add an NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume

   For more information, see the *Dell PowerFlex 5.0.x Administration Guide* and *Dell PowerFlex 5.0.x CLI Reference Guide*.
5. Discover and connect to the target PowerFlex system. See related information for more details.
6. Configure the type of multipath used:

   Multipath type: Native NVMe Multipath

   Use the following procedure:

   - Check if the native NVMe multipath is enabled using the following command:

```
cat /sys/module/nvme_core/parameters/multipath
```

   - If `Y` is returned, native NVMe multipath is already enabled.

- To enable the NVMe multipath, run the following command:

```
grubby --update-kernel=ALL --args="nvme_core.multipath=Y"
grub2-mkconfig -o /boot/grub2/grub.cfg
reboot
```

- To enable active/active protocol which remains persistent after reboots, run the following command:

```
echo 'ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{iopolicy}="round-robin"' >
/lib/udev/rules.d/71-nvme-iopolicy.rules
```

7. Configure a persistent initiator. This initiator automatically forms NVMe connections after reboots.
   a. Create a service that runs `nvme connect-all` on reboots:

```
vi /etc/systemd/system/nvme_fabrics_persistent.service
[Unit]
Description=NVMf auto discovery service
Requires=network.target
After=systemd-modules-load.service network.target
[Service]
Type=oneshot
ExecStart=/usr/sbin/nvme connect-all
StandardOutput=journal
[Install]
WantedBy=multi-user.target timers.target
```

   b. Start a new service, and enable it:

```
systemctl start nvme_fabrics_persistent.service
systemctl enable nvme_fabrics_persistent.service
```

8. Run the following command to update the udev rule to make the recovery delay 15 seconds:

```
echo ACTION!=\"remove\", KERNELS==\"ctl\", SUBSYSTEMS==\"nvme-fabrics\",
ATTR{transport}==\"tcp\", ATTR{model}==\"powerflex\", ATTR{recovery_delay}=\"15\" >>
/etc/udev/rules.d/71-nvme-recovery-delay.rules
```

NOTE: This step is required only for SUSE Linux Enterprise Server (SLES) 15.5 version.

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Select the **Volume** checkbox and click **Provision** > **Map**.
3. Select the protocol NVMe.
4. Select the checkbox on the host that you are mapping the volume to.
5. Click **Map**.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **Storage systems** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To perform the discovery to check that there is an NVMe connection, use the following command:

   ```
   #nvme discover --transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
   trsvcid=8009
   ```

   192.168.163.17 is the NVMe over TCP controller IP address (data 1 VLAN) , and **192.168.163.251** is the NVMe over TCP host IP address (data 1 VLAN).

   ```
   Discovery Log Number of Records 4, Generation counter 2
   =====Discovery Log Entry 0======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  1
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.19
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 1======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  17
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.18
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 2======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  32
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.161.20
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 3======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  49
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.17
   eflags:  not specified
   sectype: non
   ```

5. On the NVMe over TCP client, connect the client to the PowerFlex NVMe over TCP target system.
6. Configure the `/etc/nvme/discovery.conf file` to add the parameters used in the NVMe discover command and use `-p` at the end of the command for persistent controller discovery.

   Persistent discovery controller will:

   a. Notify a host of fabric changes after initial discovery.
   b. The host retains connection to discovery controller.

NOTE: Add an entry for all the SDT IP addresses discovered in the previous step:

```
echo "--transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.18 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.19 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.161.20 --host-traddr=192.168.161.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

Where

- `--traddr=192.168.161.xx`— Data 1 IP address configured in SDT.
- `--traddr=192.168.163.xx`—Data 2 IP address configured in SDT.
- `--host-traddr=192.168.163.251`—Data 1 IP address configured in the NVMe host.
- `--host-traddr=192.168.161.251`—Data 2 IP address configured in the NVMe host.

7. Connect the NVMe over TCP host to the controller system.
   `# nvme connect-all`
8. Verify that the NVMe over TCP host can access the namespace:
   `# nvme list-subsys`

Configure NVMe initiators on hosts for VMware ESXi-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example only. See VMware documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running VMware ESXi operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 5.0.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Configure at least three vSwitches: one for management, and two for data (for example: data1 and data2)
2. Configure the VMkernel on each data vSwitch:
   a. In the command line, add the **NVMeTCP** tag to the data VMkernels. For example:

   ```
   esxcli network ip interface tag add -t NVMeTCP -i vmk1
   esxcli network ip interface tag add -t NVMeTCP -i vmk2
   ```

   b. Verify that the **NVMeTCP** tag was added:

   ```
   esxcli network ip interface tag get -i vmk1
   ```

   The returned output should include the following:

   ```
   Tags: NVMeTCP
   ```

   c. Enable NVMeTCP on the data vSwitch uplink network adapters:

   ```
   esxcli nvme fabrics enable -p TCP -d vmnic4
   esxcli nvme fabrics enable -p TCP -d vmnic5
   ```

d. Verify that NVMeTCP was enabled on the network adapters:

```
esxcli nvme adapter list
```

Output similar to the following should be returned:

```
Adapter  Adapter Qualified Name          Transport Type  Driver   Associated
Devices

-------  ------------------------------  --------------  -------
------------------

vmhba64  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic4
vmhba65  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic5
```

3. Get the ESXi NQN of each ESXi node, and make a note of each one. You need this information when configuring the NVMe targets with PowerFlex Manager.

```
esxcli nvme info get
```

Output similar to the following should be returned:

```
Host NQN:nqn.2014-08.com.xyz.lab.sio:nvme:sio-dcoe-116AW-13
```

4. Perform the following actions:
   a. Add NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume.

   For more information, see the *Dell PowerFlex 5.0.x Administration Guide* and *Dell PowerFlex 5.0.x CLI Reference Guide*.

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Select the **Volume** checkbox and click **Provision** > **Map**.
3. Select the protocol NVMe.
4. Select the checkbox on the host that you are mapping the volume to.
5. Click **Map**.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **Storage systems** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To discover and connect the NVMe over TCP Target PowerFlex system, perform the following:

   NOTE: Use the esxcli command to perform the operation, there is issue with persistent discovery when perform the steps using UI.

a. Log in to VMware ESXi server using SSH.
b. Run a discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
```

Where 6x is the first NVMe over TCP software adapter, and 6y is the second NVMe over TCP software adapter.

Where 192.168.x.x is the first IP address and 192.168.y.y is the second IP address of any one of the SDTs.

c. Connect to the PowerFlex system (append **-c** on the discovery query command).

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009 -c
```

d. Get the controller list. Verify the connected controllers:

```
#esxcli nvme controller list
```

5. Perform a rescan of the storage.
   a. In the vSphere Client, browse to a host, a cluster, a data center, or a folder that contains hosts.
   b. From the right-click menu, select **Storage** > **Rescan Storage**.
   c. Select **Scan for new Storage Devices** and click **OK**.
6. Create a new VMFS datastore.
   a. Log in to vCenter
   b. Select **Home/Inventory** and select the storage icon.
   c. In the left right, click the PowerFlex compute-only cluster, and select **Storage/New Datastore**.
   d. Select **VMFS** and click **NEXT**.
   e. Enter the name of the datastore and select a host from the list.

      You should see a new disk called NVMe TCP Disk (eui.######)
   f. Select the disk and click **NEXT**.
   g. Select **VMFS 6** and click **NEXT**.
   h. Leave the default configuration unless instructed by the customer and click **NEXT**.
   i. Review your selections and click **FINISH**.